

# Dual Selective Fusion Transformer Network for Hyperspectral Image Classification

Yichu Xu<sup>a,b</sup>, Di Wang<sup>a,b</sup>, Lefei Zhang<sup>a,b,\*</sup> and Liangpei Zhang<sup>c</sup>

<sup>a</sup> Institute of Artificial Intelligence, School of Computer Science, Wuhan University, Wuhan, 430072, PR China

<sup>b</sup> Hubei LuoJia Laboratory, Wuhan 430079, PR China

<sup>c</sup> Aerospace Information Research Institute, Henan Academy of Sciences, Zhengzhou 450046, PR China

## ARTICLE INFO

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## ABSTRACT

Transformer has achieved satisfactory results in the field of hyperspectral image (HSI) classification. However, existing Transformer models face two key challenges when dealing with HSI scenes characterized by diverse land cover types and rich spectral information: (1) A fixed receptive field overlooks the effective contextual scales required by various HSI objects; (2) invalid self-attention features in context fusion affect model performance. To address these limitations, we propose a novel Dual Selective Fusion Transformer Network (DSFormer) for HSI classification. DSFormer achieves joint spatial and spectral contextual modeling by flexibly selecting and fusing features across different receptive fields, effectively reducing unnecessary information interference by focusing on the most relevant spatial-spectral tokens. Specifically, we design a Kernel Selective Fusion Transformer Block (KSFTB) to learn an optimal receptive field by adaptively fusing spatial and spectral features across different scales, enhancing the model’s ability to accurately identify diverse HSI objects. Additionally, we introduce a Token Selective Fusion Transformer Block (TSFTB), which strategically selects and combines essential tokens during the spatial-spectral self-attention fusion process to capture the most crucial contexts. Extensive experiments conducted on four benchmark HSI datasets demonstrate that the proposed DSFormer significantly improves land cover classification accuracy, outperforming existing state-of-the-art methods. Specifically, DSFormer achieves overall accuracies of 96.59%, 97.66%, 95.17%, and 94.59% in the Pavia University, Houston, Indian Pines, and Whu-HongHu datasets, respectively, reflecting improvements of 3.19%, 1.14%, 0.91%, and 2.80% over the previous model. The code will be available online at <https://github.com/YichuXu/DSFormer>.

## 1. Introduction

With the continuous advancements in hyperspectral imaging technology, hyperspectral images (HSIs) now offer increasingly rich spatial-spectral information, enabling precise Earth Observation (Yang et al., 2024; Zhang et al., 2022; Paoletti et al., 2019; Xu et al., 2023; Wang et al., 2024). Leveraging their distinctive spectral characteristics, HSIs are capable of performing a wide range of tasks, including classification (Shi et al., 2023; Li et al., 2019a; Gao et al., 2024; Roy et al., 2023a), target detection (Zhu et al., 2023; Xu et al., 2022; Xie et al., 2021), change detection (Chang and Ghamisi, 2023; Hu et al., 2023; Chen et al., 2024), and image quality enhancement (Xiao et al., 2024a; Wang et al., 2022b; Chen et al., 2023a). Notably, HSI classification has become a fundamental task in the remote sensing community due to its extensive applications in areas such as disaster monitoring (Hong et al., 2021), precision agriculture (Zhang et al., 2016), and urban planning (van der Linden and Hostert, 2009). The primary goal of HSI classification is to assign specific class labels to each pixel within the HSI.

HSI classification increasingly relies on deep learning techniques, which offer significant advantages over traditional machine learning methods due to their powerful feature extraction capabilities (Zhao and Du, 2016b; Mou et al., 2018). In particularly, convolutional neural networks (CNNs) enable the extraction of features in a hierarchical

manner (Roy et al., 2020; Xu et al., 2018a). However, their local convolutional structure limits their ability to capture long-range dependencies. To address this, recent studies (Wang et al., 2018) introduced self-attention mechanisms to capture non-local contextual information, though at a high computational cost due to extensive similarity calculations.

Recently, the success of self-attention in natural language processing (Vaswani et al., 2017) has inspired its application in computer vision. Dosovitskiy *et al.* (Dosovitskiy et al., 2021) proposed the Vision Transformer (ViT), which segments images into tokens and applies self-attention to capture long-range dependencies, achieving notable improvements in image classification. Transformer-based approaches (He et al., 2024; Zhong et al., 2022; Hong et al., 2022; Zhao et al., 2024) have become a mainstream trend in HSI classification research, with numerous studies introducing novel methods to advance HSI classification technology.

However, most transformer structures typically utilize a fixed receptive field for feature extraction, overlooking the fact that different types of land cover require varying contextual information (Li et al., 2019b, 2023). As shown in Fig. 1 (a), HSI with a fixed receptive field may exhibit misclassification issues. Therefore, we argue that different objects in HSIs should be recognized with correspondingly required receptive fields. After determining the scope of visual perception, the next step is to perform effective context capturing. To achieve this, most transformer-based HSI classification models usually employ vanilla Multi-Head Self-Attention (MHSA) to build global features (Mei et al.,

# 双重选择性融合Transformer网络用于高光谱图像分类

徐一初<sup>a,b</sup>, 王迪<sup>a,b</sup>, 张乐飞<sup>a,b,\*</sup> 和张亮培<sup>c</sup>

<sup>a</sup> 人工智能研究所, 计算机科学学院, 武汉大学, 武汉, 430072, 中国

<sup>b</sup> 湖北洛嘉实验室, 武汉430079, 中国

<sup>c</sup> 航天信息研究院, 河南省科学院, 郑州450046, 中国

## ARTICLE INFO

**关键词:** 高光谱图像分类  
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## ABSTRACT

Transformer 在高光谱图像 (HSI) 分类领域已取得令人满意的结果。然而, 现有的 Transformer 模型在处理具有多样地物覆盖类型和丰富光谱信息的高光谱场景时, 面临两个关键挑战: (1) 固定的感受野忽略了不同 HSI 对象所需的有效上下文尺度; (2) 上下文融合中的无效自注意力特征影响了模型性能。为解决这些局限性, 我们提出了一种用于 HSI 分类的新型双重选择性融合 Transformer 网络 (DSFormer)。DSFormer 通过灵活选择和融合不同感受野的特征, 实现时空联合上下文建模, 通过聚焦最相关的时空频标记, 有效减少不必要的干扰信息。具体而言, 我们设计了一个内核选择性融合 Transformer 模块 (KSFTB), 通过自适应融合不同尺度的时空和光谱特征, 学习最优感受野, 增强模型准确识别多样 HSI 对象的能力。此外, 我们引入了一个 token 选择性融合 Transformer 模块 (TSFTB), 在时空自注意力融合过程中策略性地选择和组合关键 token, 以捕获最重要的上下文。在四个基准高光谱数据集上的大量实验表明, 所提出的 DSFormer 显著提高了地物覆盖分类精度, 优于现有最先进的方法。具体而言, DSFormer 在帕维亚大学、休斯顿、印第安松林和吴红胡数据集上的总体精度分别为 96.59%、97.66%、95.17% 和 94.59%, 相较于前一个模型分别提高了 3.19%、1.14%、0.91% 和 2.80%。代码将在 <https://github.com/YichuXu/DSFormer> 上提供。

## 1. 简介

随着高光谱成像技术的不断进步, 高光谱图像 (HSIs) 现在提供日益丰富的时空谱信息, 使地球观测 (Yang等人, 2024年; Zhang等人, 2022年; Paoletti等人, 2019年; Xu等人, 2023年; Wang等人, 2024年) 成为可能。利用其独特的光谱特性, HSIs能够执行多种任务, 包括分类 (Shi等人, 2023年; Li等人, 2019a; Gao等人, 2024年; Roy等人, 2023a)、目标检测 (Zhu等人, 2023年; Xu等人, 2022年; Xie等人, 2021年)、变化检测 (Chang和Ghamisi, 2023年; Hu等人, 2023年; Chen等人, 2024年) 和图像质量增强 (Xiao等人, 2024a; Wang等人, 2022b; Chen等人, 2023a)。值得注意的是, HSI分类已成为遥感领域的一项基本任务, 因为它在灾害监测 (Hong等人, 2021年)、精准农业 (Zhang等人, 2016年) 和城市规划 (van der Linden和Hostert, 2009年) 等领域具有广泛的应用。HSI分类的主要目标是为HSI中的每个像素分配特定的类别标签。

高光谱图像分类越来越多地依赖于深度学习技术, 由于其强大的特征提取能力 (Zhao and Du, 2016b; Mou et al., 2018), 这些技术比传统的机器学习方法具有显著的优势。特别是在此, 卷积神经网络 (CNNs) 以层次化的方式提取特征 (Roy et al., 2020; Xu et al., 2018a)。然而, 它们的局部卷积结构限制了它们捕获长距离依赖的能力。为了解决这个问题, 最近的研究 (Wang et al., 2018) 引入了自注意力机制来捕获非局部上下文信息, 但由于大量的相似性计算, 这导致了高计算成本。

\*通讯作者。

尽管如此, 它们的局部卷积结构限制了它们捕获长距离依赖的能力。为了解决这个问题, 最近的研究 (Wang et al., 2018) 引入了自注意力机制来捕获非局部上下文信息, 尽管由于大量的相似性计算, 这导致了高计算成本。

近年来, 自注意力机制在自然语言处理 (Vaswani等人, 2017年) 中的成功应用, 激发了其在计算机视觉领域的应用。Dosovitskiy等人 (Dosovitskiy等人, 2021年) 提出了视觉 Transformer (ViT), 该模型将图像分割成标记, 并应用自注意力机制来捕获长距离依赖关系, 在图像分类任务中取得了显著改进。基于Transformer的方法 (He等人, 2024年; Zhong等人, 2022年; Hong等人, 2022年; Zhao等人, 2024年) 已成为高光谱图像分类研究的主流趋势, 大量研究引入了新方法以推动高光谱图像分类技术的发展。

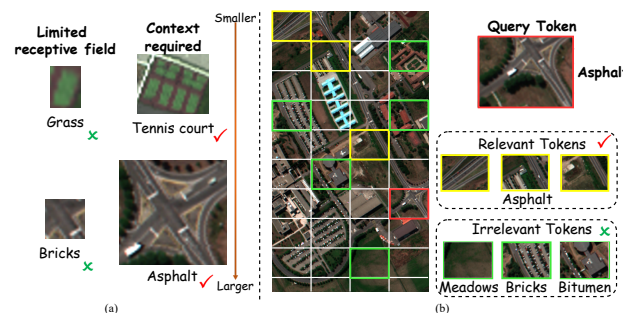
然而, 大多数Transformer结构通常使用固定的感受野进行特征提取, 忽略了不同类型的地物覆盖需要不同的上下文信息 (李等人, 2019b, 2023)。如图1 (a) 所示, 具有固定感受野的高光谱图像 (HSI) 可能会出现错分类问题。因此, 我们主张HSI中的不同对象应该使用相应所需的感受野进行识别。在确定视觉感知范围后, 下一步是进行有效的上下文捕获。为了实现这一点, 大多数基于Transformer的HSI分类模型通常采用纯多头自注意力 (MHSA) 来构建全局特征 (梅等人, 2022), 通过使一个标记与其他标记之间进行交互。然而, 由于并非所有标记都是信息性的, 原始的MHSA在上下文特征提取过程中不可避免地引入了意外的噪声。如图1 (b) 所示, 绿色框表示相对于沥青红色查询标记的不必要标记, 例如草地和沥青。此外, 在捕获上下文信息时, 现有的基于Transformer的HSI分类模型主要关注空间或光谱信息中的任何一种 (Hong等人, 2022), 忽略了HSI数据中空间-光谱上下文的集成建模。



2022) by enabling interactions between one token and all the other tokens. However, since not all tokens are informative, the original MHSA inevitably introduces unexpected noise during context feature extraction. As shown in Fig. 1 (b), the green boxes represent unnecessary tokens relative to the red query marker for asphalt, such as meadows and bitumen. Additionally, in capturing contextual information, existing transformer-based HSI classification models predominantly focus on either spatial or spectral information alone (Hong et al., 2022), neglecting the integrated modeling of spatial-spectral contexts within HSI data.

To address these issues, this paper proposes a novel Dual Selective Fusion Transformer Network (DSFormer) for HSI classification, which selects and fuses the most valuable and relevant spatial-spectral information within a reasonable receptive field. To accommodate this, DSFormer incorporates a Kernel Selective Fusion Transformer Block (KSFTB), which dynamically selects and integrates multiple convolutional kernels of appropriate sizes, optimizing the receptive fields to obtain effective spatial-spectral features. Furthermore, rather than utilizing all tokens for dense attention calculation, DSFormer introduces a Token Selective Fusion Transformer Block (TSFTB). This block selects the most relevant tokens of the highest attention values for fusion through a spectral grouping strategy that can directly utilize the 3D HSI characteristic. These selective mechanisms allow DSFormer to produce the most significant and appropriate spatial-spectral information within an optimal receptive field. The main contributions of this study are summarized as follows:

- (1) This paper proposes a novel Dual Selective Fusion Transformer Network (DSFormer) for HSI classification. It adaptively selects and fuses features from diverse receptive fields to achieve joint spatial-spectral context modeling, while reducing unnecessary information interference by focusing on the most relevant spatial-spectral tokens. This approach enables the precise classification of HSI ground objects.
- (2) To adaptively select and integrate the appropriate contextual information, we design a Kernel Selective Fusion Transformer Block (KSFTB). This mechanism ensures that features are acquired within an optimal receptive field from both spatial and spectral perspectives, thereby enhancing the model's capacity to effectively perceive and distinguish various HSI objects.
- (3) We further develop a Token Selective Fusion Transformer Block (TSFTB) that strategically selects the most relevant tokens in the context fusion process, effectively integrating essential information while reducing the impact of irrelevant objects, thereby improving the distinguishability of the generated spatial-spectral features for HSI visual interpretation.
- (4) Extensive experiments are conducted on four benchmark HSI datasets: Pavia University, Houston, Indian Pines, and Whu-HongHu. The results indicate that the



**Figure 1:** The challenges existing in current HSI classification methods. (a) Fixed and limited receptive fields in HSI classification can potentially lead to misclassifications, such as confusing asphalt to bricks. (b) Irrelevant tokens may introduce noise which may potentially impair classification accuracy.

proposed DSFormer, which utilizes appropriate and valuable contextual information, performs better than other state-of-the-art methods.

The remaining section of this paper is organized as follows. Section 2 reviews the related work of HSI classification. In Section 3 provides a comprehensive description of the proposed DSFormer model. Section 4 presents and thoroughly analyzes the experimental results. Finally, Section 5 concludes the paper.

## 2. Related Works

### 2.1. CNN-based Methods

With the rapid progress in deep learning, CNNs have shown remarkable potential and significant practical value across a wide range of computer vision tasks, particularly in the field of HSI classification. Owing to their powerful feature representation capabilities, CNNs (Zhao and Du, 2016a; Zhu et al., 2017; Chen et al., 2016; Xu et al., 2020) have attracted significant attention. A 1D-CNN (Hu et al., 2015) was proposed to achieve efficient local spectral feature extraction through multiple convolutional layers. Further, 2D-CNN was introduced in (Zhao and Du, 2016b) that extracts spatial-spectral features and uses a balanced local discriminant embedding. Later, 3D-CNNs (Ben Hamida et al., 2018; Li et al., 2017) were developed to directly extract joint spatial-spectral features from the HSI cube. Then, some multiscale convolutional networks (Xu et al., 2018b; Gong et al., 2019; Wang et al., 2021; Yang et al., 2021) were proposed to extract deep multiscale features.

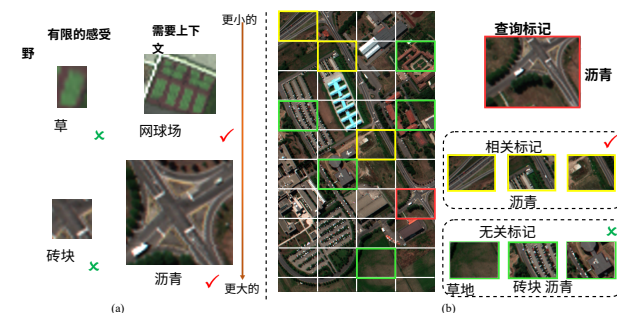
### 2.2. Transformer-based Methods

However, the above CNN-based networks are constrained by their inherent local perception, which limits the amount of information they can capture, and thus hinders accuracy. Subsequently, HSI classification approaches with attention mechanism (Sun et al., 2020; Hang et al., 2021; Zhu et al., 2021; Wang et al., 2022a) emerged and achieved significant classification results, particularly self-attention

然而, 由于并非所有标记都是信息性的, 原始的MHSA在上下文特征提取过程中不可避免地引入了意外的噪声。如图1 (b) 所示, 绿色框表示相对于沥青红色查询标记的不必要标记, 例如草地和沥青。此外, 在捕获上下文信息时, 现有的基于Transformer的HSI分类模型主要关注空间或光谱信息中的任何一种 (Hong等人, 2022), 忽略了HSI数据中空间-光谱上下文的集成建模。

为解决这些问题, 本文提出了一种用于HSI分类的新型双重选择性融合Transformer网络 (DSFormer), 该网络在合理的感受野内选择并融合最有价值和相关的时空谱信息。为了适应这一点, DSFormer集成了内核选择性融合Transformer模块 (KSFTB), 该模块动态选择并整合多个适当大小的卷积核, 优化感受野以获得有效的时空谱特征。此外, DSFormer并没有使用所有标记进行密集注意力计算, 而是引入了token选择性融合Transformer模块 (TSFTB)。该模块通过可以直接利用三维高光谱图像特征的频谱分组策略, 选择最高注意力值的最大相关标记进行融合。这些选择性机制使DSFormer能够在最优感受野内产生最显著和适当的时空谱信息。本研究的主要贡献总结如下:

- (1) 本文提出了一种用于HSI分类的新型双重选择性融合Transformer网络 (DSFormer)。它自适应地选择并融合来自不同感受野的特征, 以实现联合时空谱上下文建模, 同时通过专注于最相关的时空谱标记来减少不必要的信息干扰。这种方法能够对HSI地面物体进行精确分类。
- (2) 为了自适应地选择和整合适当的上下文信息, 我们设计了一个KernelSelectiveFusion Transformer Block (KSFTB)。该机制确保从空间和光谱角度获取最优感受野内的特征, 从而增强模型有效感知和区分各种高光谱图像 (HSI) 对象的能力。
- (3) 我们进一步开发了一个Token Selective Fusion Transformer Block (TSFTB), 该模块在上下文融合过程中策略性地选择最相关的标记, 有效整合关键信息, 同时减少无关对象的影响, 从而提高生成的时空谱特征对高光谱图像视觉解释的可区分性。
- (4) 在四个试验台上进行了大量实验  
标记高光谱数据集: 帕维亚大学、休斯顿、印第安松林和吴红胡。结果表明:



**图1:** 当前高光谱图像分类方法中存在的挑战。(a) 高光谱图像分类中固定的和有限的感受野可能导致误分类, 例如将沥青误分为砖块。(b) 无关标记可能引入噪声, 从而可能降低分类精度。

提出的 DSFormer, 该模型利用了适当且有价值上下文信息, 其性能优于其他最先进的方法。

本文剩余部分的结构安排如下。第 2 节回顾了高光谱图像分类的相关工作。第 3 节提供了对所提出 DSFormer 模型的全面描述。第 4 节展示并深入分析了实验结果。最后, 第 5 节总结了全文。

## 2. 相关工作

### 2.1. 基于CNN的方法

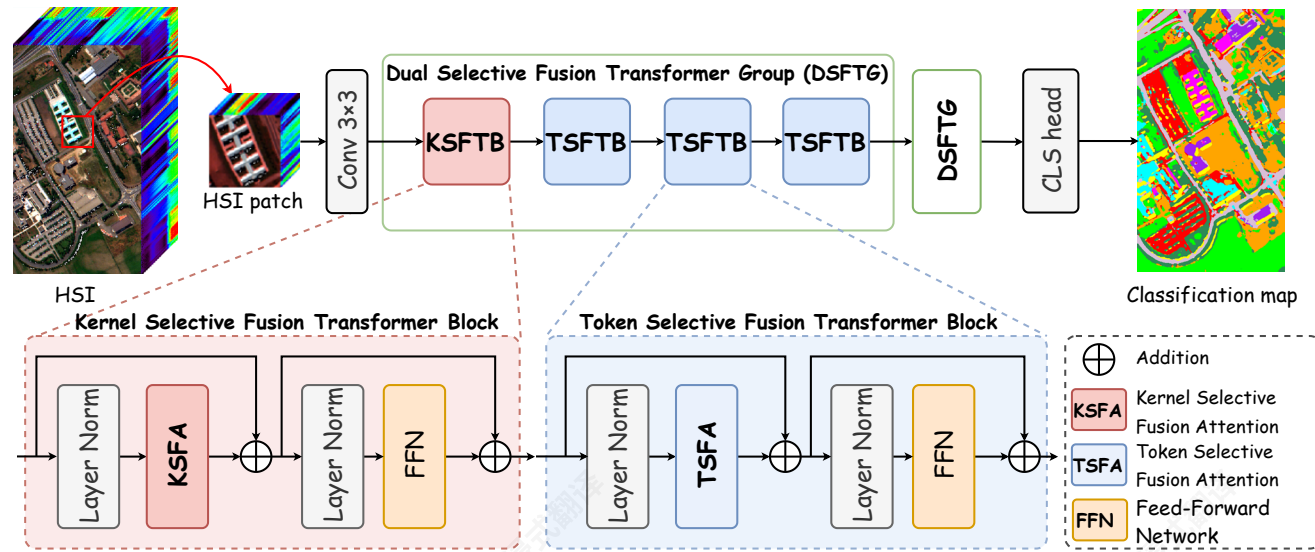
随着深度学习的快速发展, 卷积神经网络 (CNNs) 在计算机视觉任务中展现出显著的潜力和实用价值, 特别是在高光谱图像分类领域。由于强大的特征表示能力, 卷积神经网络 (Zhao and Du, 2016a; 朱等人, 2017; Chen et al., 2016; 徐等人, 2020) 引起了广泛关注。提出了一种一维卷积神经网络 (1D-CNN) (Hu et al., 2015) 以实现高效的局部光谱特征

提取。ion through multiple 对话 lutional layers. Furt h er, 二维卷积神经网络 (Zhao and Du, 2016b) 被引入, 用于提取时空谱特征并使用平衡局部判别嵌入。后来, 3D卷积神经网络 (Ben Hamida等人, 2018; 李等人, 2017) 被开发出来, 用于直接从HSI立方体中提取联合时空谱特征。然后, 一些多尺度卷积神经网络 (Xu等人, 2018b; 龚等人, 2019; 王等人, 2021; 杨等人, 2021) 被提出, 用于提取深度多尺度特征。

### 2.2. 基于Transformer的方法

然而, 上述基于CNN的网络受其固有的局部感知所限制, 这限制了它们能够捕获的信息量, 从而降低了准确率。随后, 具有注意力机制的高光谱图像分类方法 (Sun等人, 2020; Hang等人, 2021; Zhu等人, 2021; Wang等人, 2022a) 出现并取得了显著的分类结果, 尤其是自注意力





**Figure 2:** An illustration of the proposed DSFormer. The Dual Selective Fusion Transformer Group (DSFTG) is composed of a Kernel Selective Fusion Transformer Block (KSFTB) and three consecutive Token Selective Fusion Transformer Blocks (TSFTBs). Followed by two DSFTGs, we adopt the fully connected layer as the classification head to obtain the classification results.

mechanisms (Vaswani et al., 2017; Xu et al., 2021, 2024), due to their ability to improve long-range perception.

Transformer-based networks (Chen et al., 2023b; Xiao et al., 2024b; Sun et al., 2022; Cheng et al., 2024; Sun et al., 2024) have been widely adopted for their use of MHSA to achieve superior context modeling. These approaches have notably outperformed traditional CNNs, which are limited by their receptive fields. SpectralFormer was proposed in (Hong et al., 2022), a network that enhances HSI classification by capturing spectral sequence information between adjacent HSI bands using transformers and cross-layer skip connections. Further, Sun *et al.* (Sun et al., 2022) proposed a spectral-spatial feature tokenization transformer (SSFTT), where shallow features are extracted by convolutions, semantic features are processed by a Gaussian weighted tokenization module, and a single-layer transformer is used for feature representation. Then, LESSFormer was introduced (Zou et al., 2022), which employs an HSI2Token module to extract adaptive spatial-spectral tokens and uses a local transformer to enhance the local spectral representation. Inspired by the effectiveness of spectral partitioning, Mei introduced a grouped pixel embedding strategy and hierarchical structure to extract discriminative multiscale spatial-spectral features (Mei et al., 2022). Yang *et al.* (Yang et al., 2023) introduced a transformer-based center-to-surround interactive learning (CSIL) framework for extracting multiscale features for HSI classification tasks. Roy *et al.* (Roy et al., 2023b) incorporated morphological characteristics into their proposed MorphFormer by combining spectral and spatial morphological convolution operations with attention mechanisms to effectively integrate structural and morphological information. After that, Zhao *et al.* (Zhao et al., 2024) introduced a lightweight variant of the ViT network, which combines group-separable convolution and MHSA modules within feature extraction blocks to efficiently capture local

and global spatial features with reduced parameter complexity.

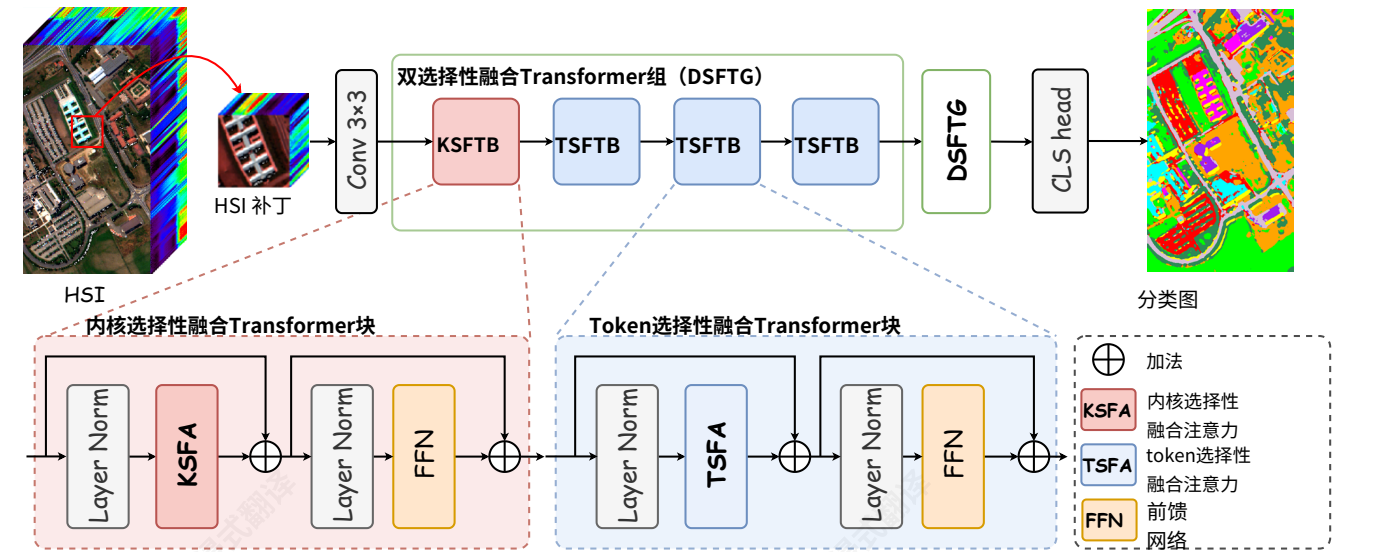
Compared with the aforementioned Transformer-based methods, we contend that the proposed DSFormer is distinct. First, KSFTB is specifically designed to adaptively choose and fuse relevant context scopes. This mechanism ensures that feature extraction takes place within an optimal receptive field, encompassing both spatial and spectral dimensions. We further develop TSFTB, which aims to combine the most relevant tokens to capture valuable information. This mechanism effectively integrates essential information while mitigating the influence of redundant and irrelevant ingredients, thereby enhancing the representation capability of spatial-spectral features for HSI classification.

### 3. Methodology

In this section, we present the implementation details of our DSFormer. DSFormer consists of convolutional layers, dual selective fusion transformer groups (DSFTGs), and a classification head. We begin with an overview of DSFormer, followed by a detailed explanation of two components KSFTB, and TSFTB.

#### 3.1. Overview of DSFormer

As shown in Fig. 2, the HSI is first subjected to PCA for dimensionality reduction, followed by the extraction of overlapping patch cubes,  $\{x_i \in \mathbb{R}^{c \times w \times w}\}_{i=1}^{H \times W}$ , where each patch is classified based on the central pixel. Here,  $H$  and  $W$  denote the height and width,  $w$  represents the patch size, and  $c$  is the reduced spectral dimension. Taking the pixel-level patch cube  $x \in \mathbb{R}^{c \times w \times w}$  as input, we first employ a  $3 \times 3$  convolutional layer to capture shallow features, denoted as  $F_1$ . Next, we perform deep feature generation on  $F_1$  using two consecutive DSFTGs. This process can be formulated



**图2:** 所提出的DSFormer的示意图。双选择性融合Transformer组 (DSFTG) 由一个内核选择性融合Transformer块 (KSFTB) 和三个连续的token选择性融合Transformer块 (TSFTBs) 组成。随后是两个DSFTG，我们采用全连接层作为分类头以获得分类结果。

机制 (Vaswani等人, 2017; Xu等人, 2021, 2024), 由于其能够提升长距离感知的能力。

基于Transformer的网络 (Chen等人, 2023b; Xiao等人, 2024b; Sun等人, 2022; Cheng等人, 2024; Sun等人, 2024) 已被广泛采用, 因为它们使用MHSA来实现卓越的上下文建模。这些方法显著优于受感受野限制的传统CNN。SpectralFormer是在 (Hong等人, 2022) 中提出的, 这是一个通过使用Transformer和跨层跳跃连接捕获相邻HSI波段之间光谱序列信息的网络, 从而增强HSI分类的网络。此外, Sun等人 (Sun等人, 2022) 提出了一种光谱-空间特征标记Transformer (SSFTT), 其中浅层特征通过卷积提取, 语义特征通过高斯加权标记化模块处理, 并使用单层Transformer进行特征表示。然后, LESSFormer被引入 (Zou等人, 2022), 它采用HSI2Token模块提取自适应时空频标记, 并使用局部Transformer增强局部光谱表示。受光谱分区的有效性的启发, Mei引入了一种分组像素嵌入策略和分层结构来提取判别性多尺度时空频特征 (Mei等人, 2022)。Yang等人 (Yang等人, 2023) 引入了一种基于Transformer的中心到周围交互学习 (CSIL) 框架, 用于提取HSI分类任务的多尺度特征。Roy等人 (Roy等人, 2023b) 通过结合光谱和空间形态学卷积操作与注意力机制, 将其提出的MorphFormer中加入了形态学特征, 以有效地整合结构和形态学信息。之后, Zhao等人 (Zhao等人, 2024) 引入了ViT网络的一个轻量级变体, 该变体在特征提取块中结合了分组可分离卷积和MHSA模块, 以高效地捕获局部

和全局空间特征, 同时降低了参数复杂度。

与上述基于Transformer的方法相比, 我们主张所提出的DSFormer具有独特性。首先, KSFTB被专门设计为自适应地选择和融合相关上下文范围。这种机制确保特征提取发生在最佳感受野内, 涵盖空间和光谱维度。我们进一步开发了TSFTB, 旨在结合最相关的token以捕获有价值的信息。这种机制有效地整合了重要信息, 同时减轻了冗余和不相关成分的影响, 从而增强了时空谱特征对高光谱图像分类的表征能力。

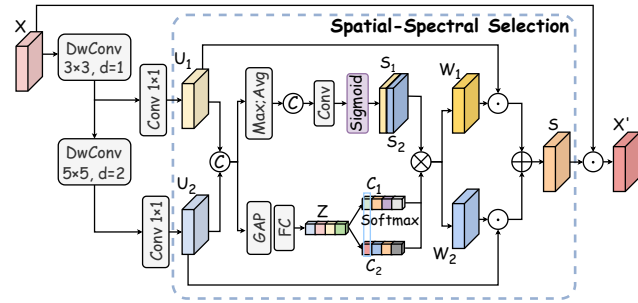
### 3. 方法论

在本节中, 我们介绍了我们的DSFormer的实现细节。DSFormer由卷积层、双选择性融合Transformer组 (DSFTGs) 和一个分类头组成。我们从DSFormer的概述开始, 然后详细解释两个组件KSFTB和TSFTB。

#### 3.1. DSFormer 概述

如图2所示, HSI首先进行PCA降维, 然后提取重叠块  $\{x_i \in \mathbb{R}^{c \times w \times w}\}_{i=1}^{H \times W}$ , 其中每个块根据中心像素进行分类。这里,  $H$  和  $W$  表示高度和宽度,  $w$  表示块大小,  $c$  是降维后的光谱维度。以像素级块  $x \in \mathbb{R}^{c \times w \times w}$  为输入, 我们首先使用一个  $3 \times 3$  卷积层来捕获浅层特征, 记为  $F_1$ 。接下来, 我们使用两个连续的DSFTG对  $F_1$  进行深度特征生成。这个过程可以表示为





**Figure 3:** The proposed Kernel Selective Fusion Attention (KSFA). Here, DwConv represents Depth-wise convolution, Avg and Max represent channel-wise average pooling and max pooling, respectively, and GAP means spatial global average pooling.

as:  $\mathbf{F}_2 = \text{DSFTG}_2(\text{DSFTG}_1(\mathbf{F}_1))$ . Finally, a layer normalization followed by a fully connected layer in the classification head is applied to obtain the final classification results.

### 3.2. Kernel Selective Fusion Transformer Block

As analyzed in the Introduction, different land cover classes in HSI require distinct contextual scales. Existing methods are often limited by the fixed and limited size of the receptive field of the network or cannot dynamically adapt the receptive field to suit specific types of land cover Sun et al. (2022); Roy et al. (2023b). To address this issue, we designed the KSFTB which learns an optimal receptive field through selecting and integrating multiscale features. KSFTB initially employs dilated depthwise convolutions to construct a large receptive field to obtain sufficient range of feature perception. Then, it incorporates a spatial-spectral selection mechanism to adaptively fusion the appropriate receptive field size required for different types of land cover. Specifically, KSFTB primarily consists of a Kernel Selective Fusion Attention (KSFA) module and a Feed-Forward Network (FFN).

The structure of KSFA is shown in Fig. 3, two decoupled depth-wise convolutions (DwConv) are employed for input  $\mathbf{X} \in \mathbb{R}^{c \times w \times w}$ . The features generated with decomposed kernels are then processed by  $1 \times 1$  convolution layers  $\mathcal{F}_{1 \times 1}(\cdot)$  to integrate channel information and unify dimensions, obtaining contextual features with different receptive fields:

$$\mathbf{U}_1 = \mathcal{F}_{1 \times 1}^1(\mathcal{F}_{3 \times 3}^{dwc}(\mathbf{X})), \mathbf{U}_2 = \mathcal{F}_{1 \times 1}^2(\mathcal{F}_{5 \times 5}^{dwc}(\mathcal{F}_{3 \times 3}^{dwc}(\mathbf{X}))), \quad (1)$$

where  $\mathcal{F}_{3 \times 3}^{dwc}$  and  $\mathcal{F}_{5 \times 5}^{dwc}$  denote a  $3 \times 3$  DwConv and a  $5 \times 5$  DwConv with a dilation of 2, respectively.

Given the varying scales of contextual information required by different types of land cover, we employ a spatial-spectral selection mechanism to enhance the dynamic adaptability of the network. Specifically, we first concatenate the features  $\mathbf{U}_1, \mathbf{U}_2 \in \mathbb{R}^{\frac{c}{2} \times w \times w}$  extracted by convolutional kernels with diverse receptive fields:

$$\mathbf{U} = [\mathbf{U}_1; \mathbf{U}_2] \in \mathbb{R}^{c \times w \times w}, \quad (2)$$

subsequently, channel-wise average pooling  $\mathcal{P}_{\text{avg}}(\cdot)$  and max pooling  $\mathcal{P}_{\text{max}}(\cdot)$  are applied to squeeze  $\mathbf{U}$  to capture spatial

descriptors. To enable information exchange among different spatial descriptors, we first concatenate the aforementioned pooling features and then using a simple  $1 \times 1$  convolutional layer  $\mathcal{F}_{1 \times 1}^{2 \rightarrow O}$  to transform the two pooling features into  $O$  spatial attention maps, we set  $O$  to 2 in our structure. These process can be written as below:

$$\tilde{\mathbf{S}} = \mathcal{F}_{1 \times 1}^{2 \rightarrow O}([\mathcal{P}_{\text{avg}}(\mathbf{U}); \mathcal{P}_{\text{max}}(\mathbf{U})]). \quad (3)$$

Following this, the sigmoid activation function  $\sigma(\cdot)$  is applied to each spatial attention feature map  $\tilde{\mathbf{S}} \in \mathbb{R}^{2 \times w \times w}$ , generating separate spatial selection masks corresponding to the different convolutional kernels:

$$\mathbf{S}_i = \sigma(\tilde{\mathbf{S}}_i). \quad (4)$$

Considering the substantial differences in contextual spectral information required by various types of complex land cover in HSIs, we have devised a spectral selection mechanism to meet this demand. First, we utilize spatial global average pooling  $\mathcal{P}_{\text{GAP}}(\cdot)$  to compress the feature  $\mathbf{U}$  into spectral feature descriptors. Next, a fully connected layer  $\mathcal{F}_{\text{FC}}(\cdot)$  is employed to produce more compact spectral attention features  $\mathbf{Z} \in \mathbb{R}^{c \times 1}$ . These operations can be mathematically expressed as follows:

$$\mathbf{Z} = \mathcal{F}_{\text{FC}}(\mathcal{P}_{\text{GAP}}(\mathbf{U})). \quad (5)$$

Subsequently, spectral-wise attention is generated through a softmax operation to guide the selection process, resulting in spectral selection masks, which can be obtained as follows:

$$\mathbf{C}_1^{[j]} = \frac{e^{M_j Z_j}}{e^{M_j Z_j} + e^{N_j Z_j}}, \mathbf{C}_2^{[j]} = \frac{e^{N_j Z_j}}{e^{M_j Z_j} + e^{N_j Z_j}}, \quad (6)$$

where  $\mathbf{C}_i^{[j]}$  represents  $j$ -th attention value in  $\mathbf{C}_i \in \mathbb{R}^{c \times 1}$ ,  $i = 1, 2$ ,  $M_j$  refers the  $j$ -th row of  $\mathbf{M}$ , likewise  $\mathbf{C}_{2j}$  and  $N_j$ , and  $\mathbf{M}, \mathbf{N} \in \mathbb{R}^{c \times 1}$  are two initially defined learnable vectors.

Furthermore, the spatial-spectral selection weights are obtained by matrix multiplication with the corresponding spatial-spectral selection masks:

$$\mathbf{W}_1 = \mathbf{C}_1 \times \mathbf{S}_1, \mathbf{W}_2 = \mathbf{C}_2 \times \mathbf{S}_2. \quad (7)$$

The enhanced feature maps derived from the decomposed kernel sequence are weighted using their respective spatial-spectral selection weights  $\mathbf{W}_i \in \mathbb{R}^{c \times w \times w}$ . These weighted maps are then combined via the convolutional layer  $\mathcal{F}_{1 \times 1}(\cdot)$ ,  $i = 1, 2$ , the attention feature  $\mathbf{S}$  can be expressed as:

$$\mathbf{S} = \mathcal{F}_{1 \times 1}(\sum_{i=1}^O (\mathbf{W}_i \cdot \mathbf{U}_i)). \quad (8)$$

The KSFA module generates its final output by performing an element-wise multiplication between the input features  $\mathbf{X}$  and  $\mathbf{S}$ :

$$\mathbf{X}' = \mathbf{X} \cdot \mathbf{S}. \quad (9)$$

The FFN module comprises a fully connected layer, a depthwise convolution, a GELU activation function, and a second fully connected layer.

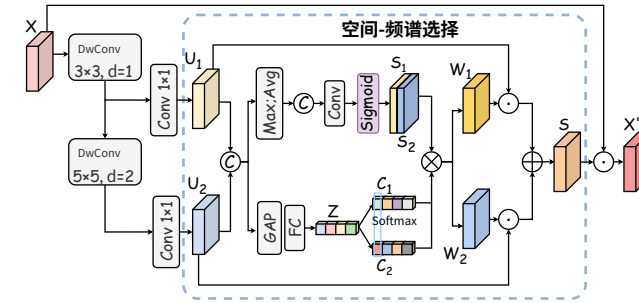


图3: 所提出的内核选择性融合注意力 (KSFA)。这里, DwConv表示深度卷积, Avg和Max分别表示通道平均池化和最大池化, GAP表示空间全局平均池化。

为:  $\mathbf{F}_2 = \text{DSFTG}_2(\text{DSFTG}_1(\mathbf{F}_1))$ 。最后, 在分类头中对层归一化后接全连接层, 以获得最终的分类结果。

### 3.2. 内核选择性融合Transformer模块

如引言中所述, HSI中的不同地物覆盖类别需要不同的上下文尺度。现有方法通常受限于网络感受野的固定和有限大小, 或无法动态调整感受野以适应特定类型的地物覆盖 (Sun等人, 2022); (Roy等人, 2023b)。为解决这个问题, 我们设计了KSFTB, 该模块通过选择和融合多尺度特征来学习最优感受野。KSFTB最初采用扩张深度卷积来构建大感受野, 以获得足够的特征感知范围。然后, 它结合了空间-频谱选择机制, 以自适应地融合不同类型地物覆盖所需的适当感受野大小。具体而言, KSFTB主要由内核选择性融合注意力 (KSFA) 模块和前馈网络 (FFN) 组成。

KSFA的结构如图3所示, 采用两个解耦的深度卷积 (DwConv) 对输入  $\mathbf{X} \in \mathbb{R}^{c \times w \times w}$  进行处理。使用分解核生成的特征随后由  $1 \times 1$  卷积层  $\mathcal{F}_{1 \times 1}(\cdot)$  进行处理, 以整合通道信息并统一尺寸, 获得具有不同感受野的上下文特征:

$$\mathbf{U}_1 = \mathcal{F}_{1 \times 1}^1(\mathcal{F}_{3 \times 3}^{dwc}(\mathbf{X})), \mathbf{U}_2 = \mathcal{F}_{1 \times 1}^2(\mathcal{F}_{5 \times 5}^{dwc}(\mathcal{F}_{3 \times 3}^{dwc}(\mathbf{X}))), \quad (1)$$

其中  $\mathcal{F}_{3 \times 3}^{dwc}$  和  $\mathcal{F}_{5 \times 5}^{dwc}$  分别表示一个  $3 \times 3$  DwConv 和一个  $5 \times 5$  DwConv, 具有2倍的膨胀率。

考虑到不同类型地物覆盖所需的上下文信息规模不同, 我们采用空间-光谱选择机制来增强网络的动态适应性。具体来说, 我们首先将使用具有不同感受野的卷积核提取的特征  $\mathbf{U}_1, \mathbf{U}_2 \in \mathbb{R}^{\frac{c}{2} \times w \times w}$  进行拼接:

$$\mathbf{U} = [\mathbf{U}_1; \mathbf{U}_2] \in \mathbb{R}^{c \times w \times w}, \quad (2)$$

随后, 对  $\mathcal{P}_{\text{avg}}(\cdot)$  进行通道平均池化  $\mathcal{P}_{\text{max}}(\cdot)$  和最大池化  $\mathbf{U}$  以压缩  $\mathbf{U}$  以捕获空间

描述符。为了实现不同空间描述符之间的信息交换, 我们首先连接上述池化特征, 然后使用一个简单的  $1 \times 1$  卷积层  $\mathcal{F}_{1 \times 1}^{2 \rightarrow O}$  将两个池化特征转换为  $O$  空间注意力图, 我们在结构中将  $O$  设置为 2。这些过程可以写为如下:

$$\tilde{\mathbf{S}} = \mathcal{F}_{1 \times 1}^{2 \rightarrow O}([\mathcal{P}_{\text{avg}}(\mathbf{U}); \mathcal{P}_{\text{max}}(\mathbf{U})]). \quad (3)$$

随后, 对每个空间注意力特征图  $\tilde{\mathbf{S}} \in \mathbb{R}^{2 \times w \times w}$  应用 Sigmoid 激活函数  $\sigma(\cdot)$ , 生成分别对应不同卷积核的空间选择掩码:

$$\mathbf{S}_i = \sigma(\tilde{\mathbf{S}}_i). \quad (4)$$

考虑到高光谱图像中各种复杂地物所需的上下文光谱信息存在显著差异, 我们设计了一种光谱选择机制以满足这一需求。首先, 我们利用空间全局平均池化  $\mathcal{P}_{\text{GAP}}(\cdot)$  将特征  $\mathbf{U}$  压缩成光谱特征描述符。接下来, 采用全连接层  $\mathcal{F}_{\text{FC}}(\cdot)$  生成更紧凑的光谱注意力特征  $\mathbf{Z} \in \mathbb{R}^{c \times 1}$ 。这些操作可以用以下数学公式表示:

$$\mathbf{Z} = \mathcal{F}_{\text{FC}}(\mathcal{P}_{\text{GAP}}(\mathbf{U})). \quad (5)$$

随后, 通过Softmax操作生成光谱级注意力, 以指导选择过程, 从而得到光谱选择掩码, 其计算方法如下:

$$\mathbf{C}_1^{[j]} = \frac{e^{M_j Z_j}}{e^{M_j Z_j} + e^{N_j Z_j}}, \mathbf{C}_2^{[j]} = \frac{e^{N_j Z_j}}{e^{M_j Z_j} + e^{N_j Z_j}}, \quad (6)$$

其中  $\mathbf{C}_i^{[j]}$  表示  $j$ -th 注意力值在  $\mathbf{C}_i \in \mathbb{R}^{c \times 1}$   $i = 1, 2$ ,  $M_j$  指代  $j$ -th 行的  $\mathbf{M}$ , 同理  $\mathbf{C}_{2j}$  和  $N_j$ , 而  $\mathbf{M}, \mathbf{N} \in \mathbb{R}^{c \times 1}$  是两个初始定义的可学习向量。

此外, 空间-频谱选择权重通过与对应的空间-频谱选择掩码进行矩阵乘法获得:

$$\mathbf{W}_1 = \mathbf{C}_1 \times \mathbf{S}_1, \mathbf{W}_2 = \mathbf{C}_2 \times \mathbf{S}_2. \quad (7)$$

从分解的核序列中得到的增强特征图使用其各自的空间-频谱选择权重  $\mathbf{W}_i \in \mathbb{R}^{c \times w \times w}$  进行加权。这些加权的特征图随后通过卷积层  $\mathcal{F}_{1 \times 1}(\cdot)$   $i = 1, 2$  进行组合, 注意力特征  $\mathbf{S}$  可以表示为:

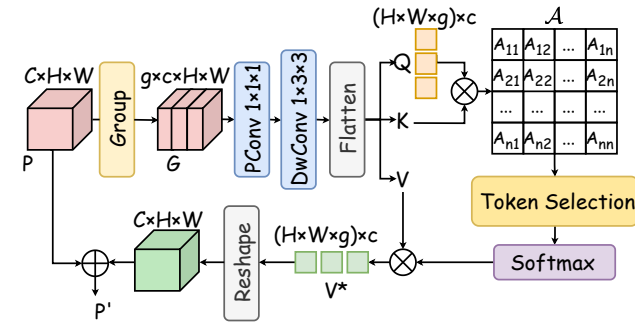
$$\mathbf{S} = \mathcal{F}_{1 \times 1}(\sum_{i=1}^O (\mathbf{W}_i \cdot \mathbf{U}_i)). \quad (8)$$

KSFA 模块通过输入特征  $\mathbf{X}$  和  $\mathbf{S}$  之间的逐元素乘法生成其最终输出:

$$\mathbf{X}' = \mathbf{X} \cdot \mathbf{S}. \quad (9)$$

FFN模块包含一个全连接层、一个深度卷积、一个GELU激活函数和一个第二个全连接层。





**Figure 4:** The illustration of the proposed Token Selective Fusion Attention (TSFA). The input feature  $P$  is first grouped and then fed through a 3D point-wise convolution, followed by a 3D depth-wise convolution to generate the corresponding  $\mathbf{QKV}$ . Subsequently, a token selection mechanism is utilized for further self-attention operation.

### 3.3. Token Selective Fusion Transformer Block

Previous research has typically utilized the self-attention mechanism within Transformers to capture long-range dependencies, facilitating the modeling of global spatial or spectral relationships Hong et al. (2022); Zhang et al. (2024). However, HSI often contains a wide variety of complex land cover types. The conventional Transformer architecture tends to compute attention across all query-key pairs, which inevitably leads to interactions between redundant and irrelevant information, ultimately diminishing interpretative accuracy.

To overcome this limitation, we propose a novel multi-head Token Selective Fusion Attention (TSFA) mechanism to replace the traditional MHSA mechanism. Furthermore, our TSFA retains the data characteristics of the HSI three-dimensional cube through a grouped operation and employs 3D convolutions to extract tokens with spatial and spectral properties. During the computation of self-attention, the TSFA selectively focuses on the most relevant tokens, effectively disregarding the least relevant information. This approach improves interpretative accuracy while reducing redundant computations.

As illustrated in Fig. 4, we first begin by partitioning the input feature  $\mathbf{P} \in \mathbb{R}^{C \times H \times W}$  into  $g$  groups, each producing a feature map of size  $c \times H \times W$ . Then we feed them through a 3D point-wise convolution with a kernel size of  $1 \times 1 \times 1$ , followed by a 3D depth-wise convolution with a kernel size of  $1 \times 3 \times 3$ . This process generates the corresponding query  $\mathbf{Q}$ , key  $\mathbf{K}$ , and value  $\mathbf{V}$ . To facilitate the calculation of self-attention, we reshape and transpose the tensors  $\mathbf{Q}$ ,  $\mathbf{K}$ , and  $\mathbf{V}$ , adjusting their dimensions by flattening the operation, so each has a shape of  $(H \times W \times g) \times c$ . These processes can be extended to the multi-head scenario and expressed as follows:

$$\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i = \text{Split}(\mathcal{F}_{1 \times 3 \times 3}^{\text{dwc}}(\mathcal{F}_{1 \times 1 \times 1}(\mathbf{e}_i))), \quad (10)$$

$$\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i = \text{Flatten}(\text{Reshape}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i)), \quad (11)$$

#### Algorithm 1 Token Selective Fusion Attention

**Input:** Feature  $\mathbf{P} \in \mathbb{R}^{C \times H \times W}$ , group number  $g$ , head number  $h$ , token selection rate  $k$ .

**Initialization:**  $\mathbf{e}_i \in \mathbb{R}^{d \times H \times W}$  is the  $i$ -th head of feature,  $d = \frac{C}{h}$ .  $\mathcal{M}^k \in \mathbb{R}^{n \times n}$  is initialized with  $-\infty$ ,  $n = HWg$ .  $c = \frac{d}{g}$ ,  $\lambda = \sqrt{d}$ .

```

1: for each  $\mathbf{e}_i$  do
2:    $\mathbf{G}_i \leftarrow \text{Group}(\mathbf{e}_i);$  //  $g \times c \times H \times W$ 
3:    $\mathbf{G}_i \leftarrow \text{DWConv}(\text{PConv}(\mathbf{G}_i));$  //  $g \times c \times H \times W$ 
4:    $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i \leftarrow \text{Flatten}(\text{Chunk}(\mathbf{G}_i));$  //  $HWg \times c$ 
5:    $\mathcal{A} \leftarrow \frac{\mathbf{Q}_i \mathbf{K}_i^T}{\lambda};$  // Dense Attention  $HWg \times HWg$ 
6:    $T_k^{[p]} \leftarrow \min(\text{Topk}\{\mathcal{A}_{p1}, \dots, \mathcal{A}_{pn}\}, p = 1, \dots, n);$  //  $1 \times 1$ 
7:    $\mathcal{M}_{pq}^k \leftarrow \mathcal{A}_{pq}$  if  $\mathcal{A}_{pq} > T_k^{[p]}$ ; //  $HWg \times HWg$ 
8:    $\mathcal{A}^* \leftarrow \text{Softmax}(\mathcal{M}^k);$  // Selective Attention  $HWg \times HWg$ 
9:    $\mathbf{V}_i^* \leftarrow \mathcal{A}^* \otimes \mathbf{V}_i;$  //  $HWg \times c$ 
10:   $\mathbf{V}_i^* \leftarrow \text{Reshape}(\mathbf{V}_i^*);$  //  $d \times H \times W$ 
11:   $\mathbf{P}_i' \leftarrow \mathbf{P}_i + \mathbf{V}_i^*$  //  $d \times H \times W$ 
12: end for
13:  $\mathbf{P}' \leftarrow \mathcal{F}_{1 \times 1}(\text{Concat}(\mathbf{P}_i'))$  //  $C \times H \times W$ 

```

where  $\mathbf{e}_i$  denotes the feature in  $i$ th head,  $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i \in \mathbb{R}^{HWg \times c}$  correspond to  $\mathbf{Q}$ ,  $\mathbf{K}$ , and  $\mathbf{V}$  in  $i$ th head, respectively, and  $g$  represents the number of groups. The  $\text{Split}(\cdot)$  function refers to the chunk operation performed along the channel dimension. Next, a dense attention matrix  $\mathcal{A} \in \mathbb{R}^{HWg \times HWg}$  is generated through the dot product between  $\mathbf{Q}_i$  and  $\mathbf{K}_i$ :

$$\mathcal{A} = \frac{\mathbf{Q}_i \mathbf{K}_i^T}{\lambda}, \quad (12)$$

where  $\lambda = \sqrt{\frac{c}{h}}$ ,  $h$  denotes the number of heads. Following this, the token selection mechanism is applied to identify the top  $k$  percent of the most relevant elements within  $\mathcal{A}$ . For example, with  $k = 0.8$ , the top 80% of the elements are retained for activation, while the remaining 20% are masked to 0. To achieve this, a matrix  $\mathcal{M}^k$  is created, which operates as follows:

$$\mathcal{M}_{pq}^k = \begin{cases} \mathcal{A}_{pq}, & \mathcal{A}_{pq} > T_k^{[p]} \\ -\infty, & \text{otherwise,} \end{cases} \quad (13)$$

where  $\mathcal{M}_{pq}^k$  denotes the value at position  $pq$ , and  $T_k \in \mathbb{R}^{HWg \times 1}$  is the lower threshold vector of the highest  $k$  percent attention values in each row of  $\mathcal{A}$ , i.e., each token has a unique threshold, and  $T_k^{[p]}$  is the  $p$ th value of  $T_k$ . After that, the selective attention matrix is activated by the softmax function. The activated attention can be written as follows:

$$\mathcal{A}^* = \text{Softmax}(\mathcal{M}^k), \quad (14)$$

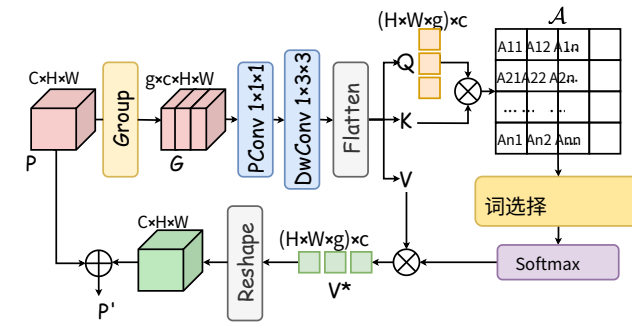


图4: 所提出的Token Selective Fusion Attention (TSFA)的示意图。输入特征  $P$  首先被分组, 然后通过3D逐点卷积, 接着通过3D深度卷积生成相应的  $\mathbf{QKV}$ 。随后, 利用token selection mechanism进行进一步的self-attention操作。

### 3.3. Token Selective Fusion Transformer Block

以往研究通常利用Transformer中的自注意力机制来捕获长距离依赖关系, 从而促进全局空间或频谱关系的建模。Hong等人(2022); 张等人(2024)。然而, HSI通常包含各种复杂的土地覆盖类型。传统Transformer架构倾向于对所有查询-键对进行注意力计算, 这不可避免地会导致冗余和不相关信息之间的交互, 最终降低解释性精度。

为了克服这一限制, 我们提出了一种新的多头Token Selective Fusion Attention (TSFA)机制来替代传统MHSA机制。此外, 我们的TSFA通过分组操作保留了HSI三维立体的数据特征, 并采用3D卷积来提取具有空间和光谱特性的标记。在自注意力计算过程中, TSFA选择性地关注最相关的标记, 有效地忽略最不相关的信息。这种方法提高了解释性精度, 同时减少了冗余计算。

如图4所示, 我们首先将输入特征  $\mathbf{P} \in \mathbb{R}^{C \times H \times W}$  分为  $g$  组, 每组生成一个大小为  $c \times H \times W$  的特征图。然后通过一个内核大小为  $1 \times 1 \times 1$  的3D逐点卷积, 接着通过一个内核大小为  $1 \times 3 \times 3$  的3D深度卷积。这个过程生成了相应的查询  $\mathbf{Q}$ 、键  $\mathbf{K}$  和值  $\mathbf{V}$ 。为了便于自注意力计算, 我们重塑和转置张量  $\mathbf{Q}$ 、 $\mathbf{K}$  和  $\mathbf{V}$ , 通过展平操作调整它们的维度, 使每个形状为  $(H \times W \times g) \times c$ 。这些过程可以扩展到多头场景, 并表达如下:

$$\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i = \text{Split}(\mathcal{F}_{1 \times 3 \times 3}^{\text{dwc}}(\mathcal{F}_{1 \times 1 \times 1}(\mathbf{e}_i))), \quad (10)$$

$$\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i = \text{Flatten}(\text{Reshape}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i)), \quad (11)$$

#### Algorithm 1 Token Selective Fusion Attention

**Input:** Feature  $\mathbf{P} \in \mathbb{R}^{C \times H \times W}$ , group number  $g$ , head number  $h$ , token selection rate  $k$ .

**Initialization:**  $\mathbf{e}_i \in \mathbb{R}^{d \times H \times W}$  is the  $i$ -th head of feature,  $d = \frac{C}{h}$ .  $\mathcal{M}^k \in \mathbb{R}^{n \times n}$  is initialized with  $-\infty$ ,  $n = HWg$ .  $c = \frac{d}{g}$ ,  $\lambda = \sqrt{d}$ .

```

1: for each  $\mathbf{e}_i$  do
2:    $\mathbf{G}_i \leftarrow \text{Group}(\mathbf{e}_i);$  //  $g \times c \times H \times W$ 
3:    $\mathbf{G}_i \leftarrow \text{DWConv}(\text{PConv}(\mathbf{G}_i));$  //  $g \times c \times H \times W$ 
4:    $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i \leftarrow \text{Flatten}(\text{Chunk}(\mathbf{G}_i));$  //  $HWg \times c$ 
5:    $\mathcal{A} \leftarrow \frac{\mathbf{Q}_i \mathbf{K}_i^T}{\lambda};$  // Dense Attention  $HWg \times HWg$ 
6:    $T_k^{[p]} \leftarrow \min(\text{Topk}\{\mathcal{A}_{p1}, \dots, \mathcal{A}_{pn}\}, p = 1, \dots, n);$  //  $1 \times 1$ 
7:    $\mathcal{M}_{pq}^k \leftarrow \mathcal{A}_{pq}$  if  $\mathcal{A}_{pq} > T_k^{[p]}$ ; //  $HWg \times HWg$ 
8:    $\mathcal{A}^* \leftarrow \text{Softmax}(\mathcal{M}^k);$  // Selective Attention  $HWg \times HWg$ 
9:    $\mathbf{V}_i^* \leftarrow \mathcal{A}^* \otimes \mathbf{V}_i;$  //  $HWg \times c$ 
10:   $\mathbf{V}_i^* \leftarrow \text{Reshape}(\mathbf{V}_i^*);$  //  $d \times H \times W$ 
11:   $\mathbf{P}_i' \leftarrow \mathbf{P}_i + \mathbf{V}_i^*$  //  $d \times H \times W$ 
12: end for
13:  $\mathbf{P}' \leftarrow \mathcal{F}_{1 \times 1}(\text{Concat}(\mathbf{P}_i'))$  //  $C \times H \times W$ 

```

其中  $\mathbf{e}_i$  表示第  $i$  个头的特征,  $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i \in \mathbb{R}^{HWg \times c}$  分别对应第  $i$  个头中的  $\mathbf{Q}$ 、 $\mathbf{K}$  和  $\mathbf{V}$ ,  $g$  表示组数。Split( $\cdot$ ) 函数指的是沿 channel 维度执行的 chunk operation。接下来, 通过  $\mathbf{Q}_i$  和  $\mathbf{K}_i$  之间的点积生成密集注意力矩阵  $\mathcal{A} \in \mathbb{R}^{HWg \times HWg}$ :

$$\mathcal{A} = \frac{\mathbf{Q}_i \mathbf{K}_i^T}{\lambda}, \quad (12)$$

其中  $\lambda = \sqrt{\frac{c}{h}}$  表示头的数量。随后, 应用标记选择机制来识别  $\mathcal{A}$  中最相关的元素的前  $k$  %。例如, 对于  $k = 0.8$ , 保留前80%的元素进行激活, 其余20%被掩码为0。为了实现这一点, 创建了一个矩阵  $\mathcal{M}^k$ , 其工作原理如下:

$$\mathcal{M}_{pq}^k = \begin{cases} \mathcal{A}_{pq}, & \mathcal{A}_{pq} > T_k^{[p]} \\ -\infty, & \text{otherwise,} \end{cases} \quad (13)$$

其中  $\mathcal{M}_{pq}^k$  表示位置  $pq$  处的值, 而  $T_k \in \mathbb{R}^{HWg \times 1}$  是  $\mathcal{A}$  每行中最高  $k$  百分比注意力值的最小阈值向量, 即每个 token 都有一个唯一阈值, 而  $T_k^{[p]}$  是  $T_k$  的第  $p$  个值。之后, 选择性注意力矩阵通过 softmax 函数被激活。激活的注意力可以表示如下:

$$\mathcal{A}^* = \text{Softmax}(\mathcal{M}^k), \quad (14)$$



**Table 1**

Performance contribution of different module on four datasets (reported in OA (%)). Best results are highlighted in **Bold**.

| Depth | KSFTB | TSFTB | Pavia University | Houston      | Indian Pines | Whu-HongHu   |
|-------|-------|-------|------------------|--------------|--------------|--------------|
| 1     | ✗     | ✗     | 87.67            | 92.51        | 85.97        | 88.32        |
| 9     | ✗     | ✗     | 86.33            | 90.04        | 85.99        | 86.58        |
| 9     | ✓     | ✗     | 93.44            | 96.98        | 92.14        | 92.95        |
| 9     | ✗     | ✓     | 95.81            | 97.44        | 93.65        | 93.84        |
| 9     | ✓     | ✓     | <b>96.59</b>     | <b>97.66</b> | <b>95.17</b> | <b>94.59</b> |

**Table 2**

The performance of spatial-spectral receptive field selective fusion mechanism on four datasets (reported in OA (%)). The best results are highlighted in **Bold**. “w/o Spa/Spe” represents the absence of the fusion of spatial or spectral receptive fields, while “w/ Spa-Spe” indicates that spatial and spectral features of different scales are employed.

| DSFormer   | Pavia University | Houston      | Indian Pines | Whu-HongHu   |
|------------|------------------|--------------|--------------|--------------|
| w/o Spa    | 96.01            | 97.41        | 94.20        | 94.13        |
| w/o Spe    | 96.09            | 97.43        | 94.38        | 94.32        |
| w/ Spa-Spe | <b>96.59</b>     | <b>97.66</b> | <b>95.17</b> | <b>94.59</b> |

Further, the  $i$ th head selective attention  $\mathbf{V}_i^*$  can be multiplied by  $\mathcal{A}^*$  and  $\mathbf{V}_i$ , and then add the input feature  $\mathbf{P}_i$  for residual connection:

$$\mathbf{P}_i' = \mathbf{P}_i + \mathcal{A}_i^* \mathbf{V}_i. \quad (15)$$

Given our multi-head design, we concatenate the outputs from each head and then aggregate them using a  $1 \times 1$  convolution, as demonstrated below:

$$\mathbf{P}' = F_{1 \times 1}(\text{Concat}(\mathbf{P}_i')). \quad (16)$$

The detailed implementation of our Token Selective Fusion Attention is presented in Algorithm 1.

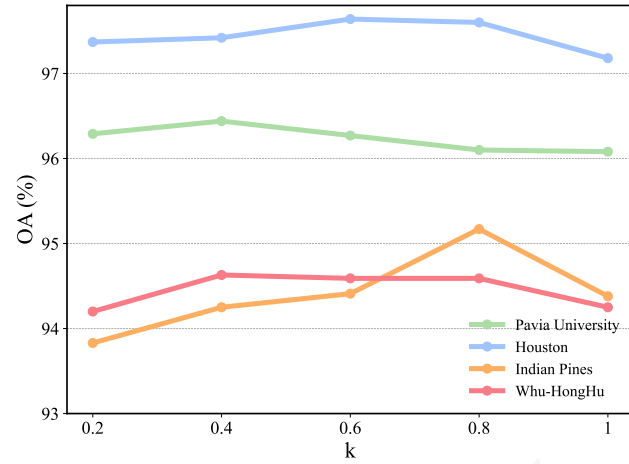
## 4. Experiment

In this section, we first provide an overview of the datasets and the implementation details. We then conduct a comprehensive evaluation of the proposed method, including in-depth parameter analysis and module ablation studies. This is followed by a qualitative and quantitative comparison of our approach with current state-of-the-art methods. Finally, we evaluated and discussed the operational efficiency of our model compared to other methods.

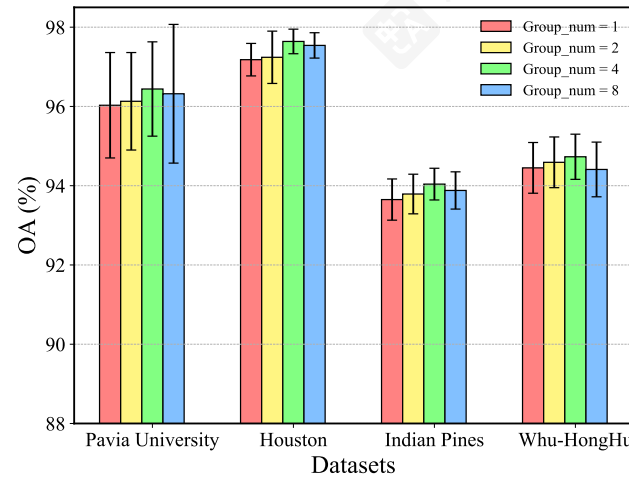
### 4.1. Data Descriptions

- (1) **Pavia University**<sup>1</sup>. This dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) on the campus of Pavia University in northern Italy in 2001. It has 103 spectral bands. The dataset consists of 610 by 340 pixels and covers 9 main categories of interest.

<sup>1</sup>[https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral\\_Remote\\_Sensing\\_Scenes](https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes)



**Figure 5:** Comparison of classification performance across four datasets under different values of  $k$ .



**Figure 6:** Comparison of classification performance across four datasets under different group number  $g$ .

- (2) **Houston**. The Houston dataset was acquired over the University of Houston campus and its surrounding areas in 2012 (Debes et al., 2014). It has 144 spectral bands in the wavelength range of 400-1000 nm and a spatial resolution of 2.5 m. It consists of 349 by 1905 pixels and covers 15 categories.
- (3) **Indian Pines**<sup>1</sup>. The AVIRIS sensor acquired this dataset in North-west Indiana in 1992. It has a dimension of 145 by 145 pixels and a spatial resolution of 20m. It covers 200 spectral bands in the wavelength range of 400-2500 nm after removing the water absorption bands. The dataset includes 16 vegetation classes.
- (4) **Whu-HongHu**. This dataset was collected using a UAV platform in HongHu City in 2017 (Zhong et al., 2020). It comprises 270 spectral bands ranging from 400 to 1000 nm, with a high spatial resolution of approximately 0.04m. The dimensions of the imagery are 940 by 475 pixels, which includes 17 typical crop types.

**表1**

不同模块在四个数据集上的性能贡献（以OA (%) 报告。最佳结果以粗体突出显示。

| 深度 | KSFTB | TSFTB | 帕维亚大学        | 休斯顿          | 印第安松林        | 吴红胡          |
|----|-------|-------|--------------|--------------|--------------|--------------|
| 1  | ✗     | ✗     | 87.67        | 92.51        | 85.97        | 88.32        |
| 9  | ✗     | ✗     | 86.33        | 90.04        | 85.99        | 86.58        |
| 9  | ✓     | ✗     | 93.44        | 96.98        | 92.14        | 92.95        |
| 9  | ✗     | ✓     | 95.81        | 97.44        | 93.65        | 93.84        |
| 9  | ✓     | ✓     | <b>96.59</b> | <b>97.66</b> | <b>95.17</b> | <b>94.59</b> |

**表2**

时空谱感受野选择性融合机制在四个数据集（报告的OA (%)）上的性能。最佳结果以粗体突出显示。“w/o Spa/Spe”表示时空感受野的融合不存在，而“w/ Spa-Spe”表示使用了不同尺度的空间和光谱特征。

| DSFormer   | 帕维亚大学        | 休斯顿          | 印第安松林        | 吴红胡          |
|------------|--------------|--------------|--------------|--------------|
| 无时空感受野     | 96.01        | 97.41        | 94.20        | 94.13        |
| w/o Spe    | 96.09        | 97.43        | 94.38        | 94.32        |
| w/ Spa-Spe | <b>96.59</b> | <b>97.66</b> | <b>95.17</b> | <b>94.59</b> |

此外，第  $i$  个头部的选择性注意力  $\mathbf{V}_i^*$  可以乘以  $\mathcal{A}^*$  和  $\mathbf{V}_i$ ，然后加上输入特征  $\mathbf{P}_i$  以进行残差连接：

$$\mathbf{P}_i' = \mathbf{P}_i + \mathcal{A}_i^* \mathbf{V}_i. \quad (15)$$

Given our 多头设计, we concatenate the outputs from each head and then aggregate them using a  $1 \times 1$  对话- lution, as demonstrated below:

$$\mathbf{P}' = F_{1 \times 1}(\text{Concat}(\mathbf{P}_i')). \quad (16)$$

The detailed implementation of our Token选择性融合注意力 is presented in Algorithm 1.

## 4. 实验

在本节中，我们首先概述数据集和实现细节。然后，我们对提出的方法进行全面评估，包括深入参数分析和模块消融研究。接下来，我们将我们的方法与当前最先进的方法进行定性和定量比较。最后，我们评估并讨论了与其他方法相比，我们模型的运行效率。

### 4.1. 数据描述

- (1) 帕维亚大学<sup>1</sup>. 该数据集由帕维亚大学在意大利北部校园的反射光学系统成像光谱仪（ROSIS）于2001年获取。它有103个光谱波段。该数据集由610×340像素组成，涵盖了15个主要感兴趣类别。

<sup>1</sup>[https://www.ehu.eus/ccwintco/index.php?title=高光谱\\_遥感\\_场景\\_](https://www.ehu.eus/ccwintco/index.php?title=高光谱_遥感_场景_)

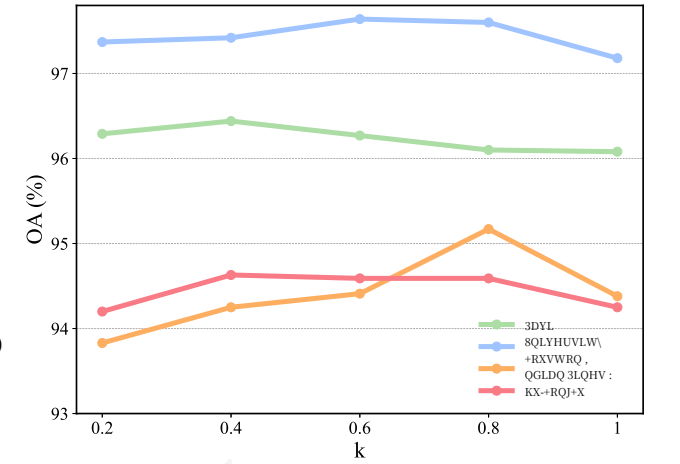


图5：在不同  $k$  值下，四个数据集的分类性能比较。

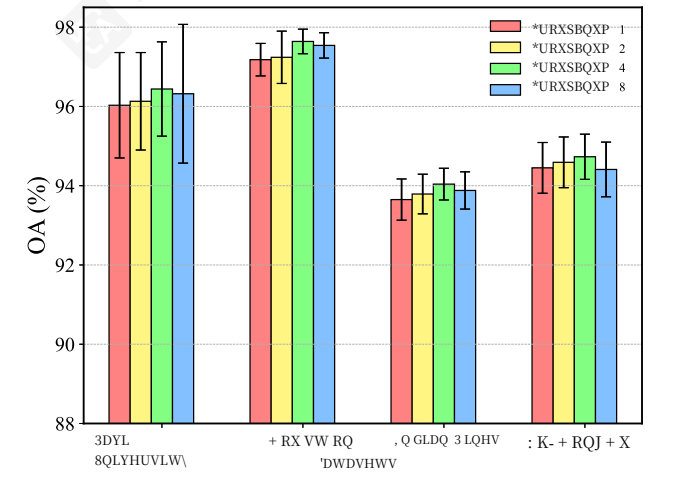


图6：在不同组数  $g$  下，四个数据集的分类性能比较。

- (2) 休斯顿. 休斯顿数据集于2012年（德贝斯等人，2014年）在休斯顿大学校园及其周边地区获取。它在400-1000纳米的波长范围内有144个光谱波段，空间分辨率为2.5米。它由349×1905像素组成，涵盖了15个类别。

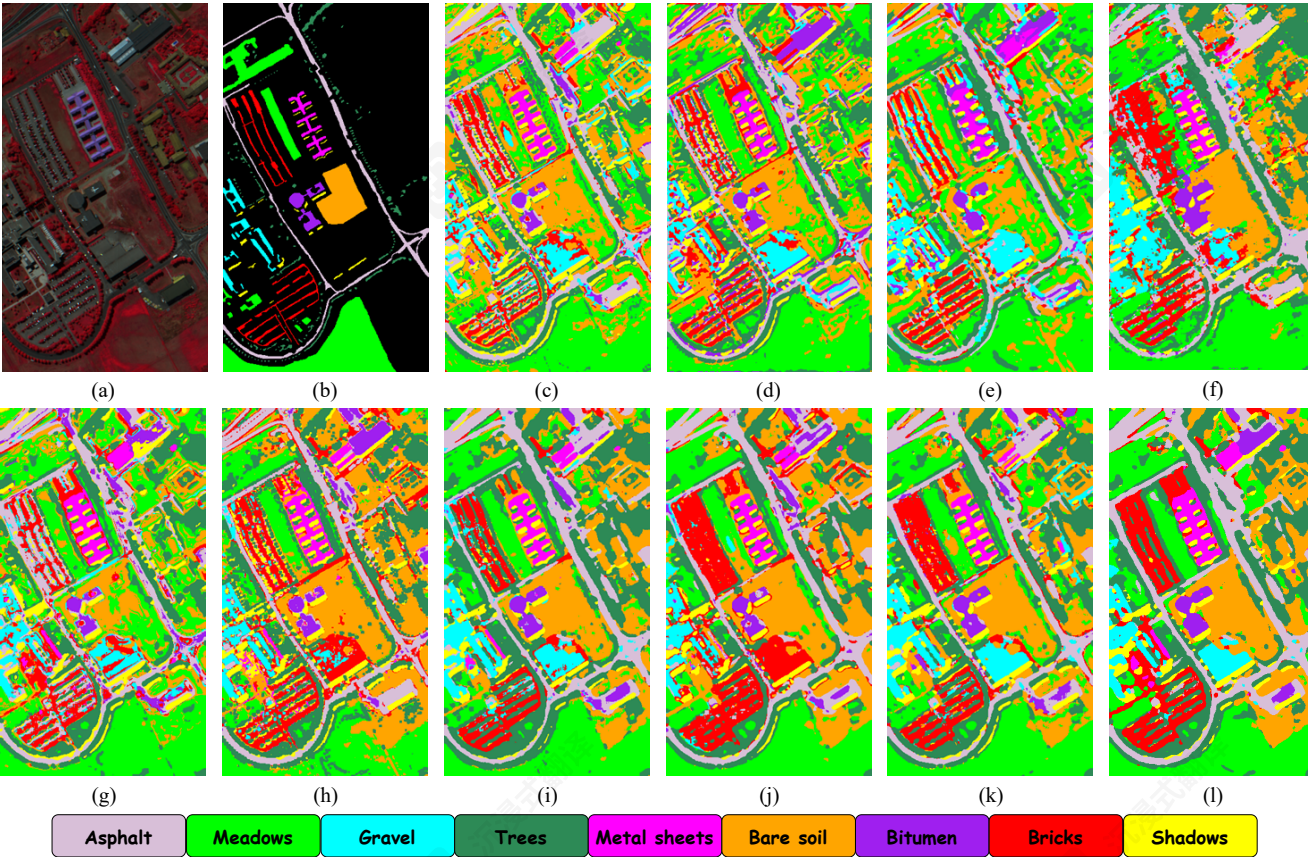
- (3) 印第安松林<sup>1</sup>. AVIRIS传感器于1992年在印第安纳州西北部获取了该数据集。其尺寸为145×145像素，空间分辨率为20米。在去除水吸收波段后，它覆盖了400-2500纳米波长范围内的200个光谱波段。该数据集包括16个植被类别。

- (4) 吴红胡. 该数据集于2017年在红湖市使用无人机平台采集（钟等人，2020）。它包含270个光谱波段，范围从400至1000纳米，具有约0.04米的高空间分辨率。影像的尺寸为940×475像素，包括17种典型作物类型。



**Table 3**  
Quantitative classification results of the Pavia University dataset. Best results are shown in **bold**.

| Class        | 3-DCNN      | SSFCN      | SACNet     | FcontNet   | SpectralFormer | SSFTT       | GAHT       | MorphFormer | GSCViT     | DSFormer          |
|--------------|-------------|------------|------------|------------|----------------|-------------|------------|-------------|------------|-------------------|
| 1            | 77.43±7.99  | 77.51±1.48 | 80.10±3.43 | 82.17±3.72 | 66.00±11.47    | 77.11±17.13 | 88.39±9.29 | 87.79±6.34  | 93.48±3.62 | <b>96.08±1.65</b> |
| 2            | 70.94±10.10 | 78.88±2.21 | 80.12±3.82 | 90.74±2.95 | 77.25±8.09     | 69.92±20.30 | 89.27±4.43 | 90.40±4.98  | 91.69±6.45 | <b>95.76±3.15</b> |
| 3            | 68.47±14.58 | 76.25±2.10 | 82.13±4.84 | 95.35±2.63 | 80.32±11.09    | 60.63±14.57 | 87.25±6.23 | 74.95±16.35 | 94.09±4.18 | <b>95.57±2.45</b> |
| 4            | 93.48±7.12  | 89.31±1.52 | 77.97±3.18 | 86.59±4.38 | 91.10±2.12     | 97.26±2.03  | 96.87±1.65 | 93.50±3.43  | 93.61±4.88 | <b>97.09±0.76</b> |
| 5            | 99.56±0.42  | 98.09±1.53 | 95.70±1.47 | 99.64±0.36 | 99.86±0.21     | 99.85±0.22  | 99.85±0.19 | 99.54±0.51  | 99.86±0.15 | <b>99.94±0.12</b> |
| 6            | 75.85±13.08 | 82.65±2.37 | 91.96±2.09 | 96.54±2.14 | 77.09±11.96    | 92.96±5.98  | 94.20±4.21 | 93.86±4.74  | 98.00±1.35 | <b>98.00±1.35</b> |
| 7            | 88.00±3.75  | 86.72±1.84 | 92.35±2.37 | 95.45±0.89 | 75.66±10.84    | 78.88±20.94 | 97.07±3.82 | 93.50±5.26  | 98.53±1.17 | <b>99.51±0.42</b> |
| 8            | 82.22±15.26 | 84.50±2.23 | 86.65±1.12 | 90.72±3.21 | 62.86±20.38    | 76.74±20.03 | 88.08±5.43 | 91.89±6.18  | 95.12±1.94 | <b>97.04±0.85</b> |
| 9            | 99.42±0.67  | 99.09±0.63 | 98.46±0.77 | 95.53±4.20 | 96.80±1.23     | 98.83±1.95  | 99.35±0.49 | 99.64±0.44  | 99.12±0.66 |                   |
| OA (%)       | 77.01±4.48  | 81.48±1.45 | 83.27±1.72 | 90.54±2.17 | 76.46±2.36     | 75.79±9.10  | 90.69±2.44 | 90.62±2.29  | 93.40±2.43 | <b>96.59±1.41</b> |
| $\kappa$ (%) | 70.95±5.22  | 76.23±1.78 | 78.53±2.03 | 87.68±2.74 | 69.94±2.64     | 69.92±9.98  | 87.91±3.09 | 87.77±2.89  | 91.39±3.05 | <b>95.52±1.82</b> |
| AA (%)       | 83.93±2.36  | 85.89±0.98 | 87.27±0.91 | 92.53±1.66 | 80.77±1.91     | 81.81±4.10  | 93.23±1.38 | 91.72±1.69  | 95.54±1.07 | <b>97.57±0.51</b> |



**Figure 7:** Visualization of the classification maps on the Pavia University dataset. (a) False-color image. (b) Ground truth. (c) 3-DCNN. (d) SSFCN. (e) SACNet. (f) FcontNet. (g) SpectralFormer. (h) SSFTT. (i) GAHT. (j) MorphFormer. (k) GSCViT. (l) DSFormer.

## 4.2. Experimental Settings

The experiments in this study were conducted on the PyTorch platform, utilizing an NVIDIA GeForce RTX 3090 GPU. For the Pavia University dataset, 30 samples per class were randomly selected for training. For the other three datasets, 50 samples per class were chosen, except for classes 1, 7, and 9 of the Indian Pines dataset, where only 15 samples were selected. The remaining labeled samples were used for testing. The training process used the AdamW optimizer, starting with an initial learning rate of  $1e-4$  and a weight decay of  $1e-5$ , was run for 500 epochs, extract patch cubes of size  $C \times 10 \times 10$ , patch size in transformer is 2,

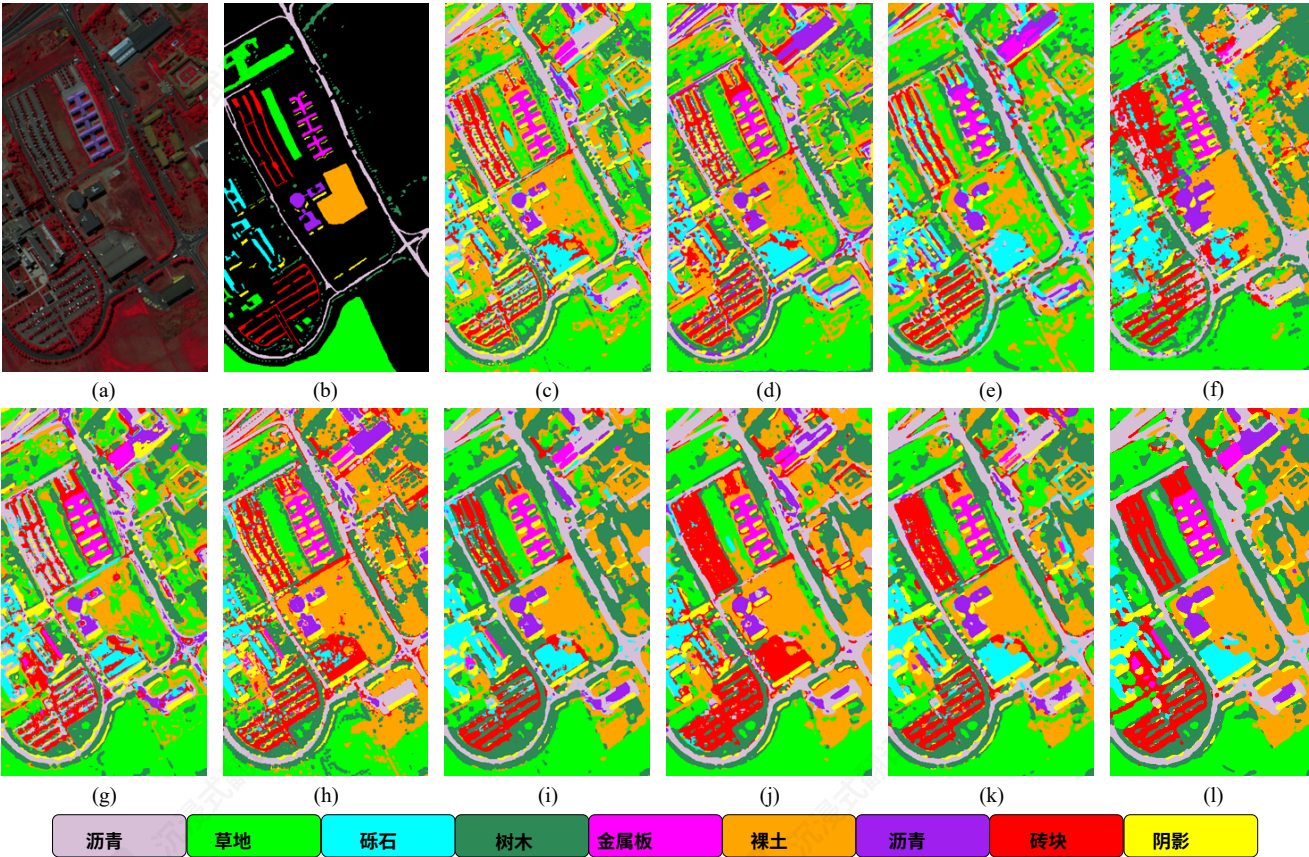
and embedding dimension is fixed with 128 in all datasets. To quantitatively assess classification performance, we used three widely recognized evaluation metrics: overall accuracy (OA), average accuracy (AA), and kappa coefficient ( $\kappa$ ). Each experiment was repeated 10 times with different random seeds, and the classification results are presented as mean and standard deviation.

## 4.3. Parameter Analysis

(1) **Effect of Different Values  $k$ .** In the TSFTB module, one of the core parameters to select the  $k$  tokens is the value of  $k$ . To identify the optimal value of  $k$ , we

**表3**  
帕维亚大学数据集的定量分类结果。最佳结果以粗体显示。

| 类别           | 3-DCNN      | SSFCN      | SACNet     | FcontNet   | SpectralFormer | SSFTT       | GAHT       | MorphFormer | GSCViT     | DSFormer          |
|--------------|-------------|------------|------------|------------|----------------|-------------|------------|-------------|------------|-------------------|
| 1            | 77.43±7.99  | 77.51±1.48 | 80.10±3.43 | 82.17±3.72 | 66.00±11.47    | 77.11±17.13 | 88.39±9.29 | 87.79±6.34  | 93.48±3.62 | <b>96.08±1.65</b> |
| 2            | 70.94±10.10 | 78.88±2.21 | 80.12±3.82 | 90.74±2.95 | 77.25±8.09     | 69.92±20.30 | 89.27±4.43 | 90.40±4.98  | 91.69±6.45 | <b>95.76±3.15</b> |
| 3            | 68.47±14.58 | 76.25±2.10 | 82.13±4.84 | 95.35±2.63 | 80.32±11.09    | 60.63±14.57 | 87.25±6.23 | 74.95±16.35 | 94.09±4.18 | <b>95.57±2.45</b> |
| 4            | 93.48±7.12  | 89.31±1.52 | 77.97±3.18 | 86.59±4.38 | 91.10±2.12     | 97.26±2.03  | 96.87±1.65 | 93.50±3.43  | 93.61±4.88 | <b>97.09±0.76</b> |
| 5            | 99.56±0.42  | 98.09±1.53 | 95.70±1.47 | 99.64±0.36 | 99.86±0.21     | 99.85±0.22  | 99.85±0.19 | 99.54±0.51  | 99.86±0.15 | <b>99.94±0.12</b> |
| 6            | 75.85±13.08 | 82.65±2.37 | 91.96±2.09 | 96.54±2.14 | 77.09±11.96    | 92.96±5.98  | 94.20±4.21 | 93.86±4.74  | 98.00±1.35 | <b>98.00±1.35</b> |
| 7            | 88.00±3.75  | 86.72±1.84 | 92.35±2.37 | 95.45±0.89 | 75.66±10.84    | 78.88±20.94 | 97.07±3.82 | 93.50±5.26  | 98.53±1.17 | <b>99.51±0.42</b> |
| 8            | 82.22±15.26 | 84.50±2.23 | 86.65±1.12 | 90.72±3.21 | 62.86±20.38    | 76.74±20.03 | 88.08±5.43 | 91.89±6.18  | 95.12±1.94 | <b>97.04±0.85</b> |
| 9            | 99.42±0.67  | 99.09±0.63 | 98.46±0.77 | 95.53±4.20 | 96.80±1.23     | 98.83±1.95  | 99.35±0.49 | 99.64±0.44  | 99.12±0.66 |                   |
| OA (%)       | 77.01±4.48  | 81.48±1.45 | 83.27±1.72 | 90.54±2.17 | 76.46±2.36     | 75.79±9.10  | 90.69±2.44 | 90.62±2.29  | 93.40±2.43 | <b>96.59±1.41</b> |
| $\kappa$ (%) | 70.95±5.22  | 76.23±1.78 | 78.53±2.03 | 87.68±2.74 | 69.94±2.64     | 69.92±9.98  | 87.91±3.09 | 87.77±2.89  | 91.39±3.05 | <b>95.52±1.82</b> |
| AA (%)       | 83.93±2.36  | 85.89±0.98 | 87.27±0.91 | 92.53±1.66 | 80.77±1.91     | 81.81±4.10  | 93.23±1.38 | 91.72±1.69  | 95.54±1.07 | <b>97.57±0.51</b> |



**图7:** 帕维亚大学数据集上的分类图可视化。(a) 假彩色图像。(b) 真实情况。(c) 3-DCNN。(d) SSFCN。(e) SACNet。(f) FcontNet。(g) SpectralFormer。(h) SSFTT。(i) GAHT。(j) MorphFormer。(k) GSCViT。(l) DSFormer。

## 4.2. 实验设置

本研究中的实验是在PyTorch平台上进行的，使用了NVIDIA GeForce RTX 3090 GPU。对于帕维亚大学数据集，每个类别随机选择30个样本进行训练。对于其他三个数据集，每个类别选择50个样本，印度针叶树数据集的类别1、7和9除外，这些类别只选择了15个样本。其余的标记样本用于测试。训练过程使用AdamW优化器，初始学习率为 $1e-4$ ，权重衰减为 $1e-5$ ，运行500轮，提取大小为 $C \times 10 \times 10$ 的重叠块，Transformer中的补丁大小为2，

并且嵌入维度在所有数据集中固定为128。为了定量评估分类性能，我们使用了三种广泛认可的评估指标：总体准确率（OA）、平均准确率（AA）和kappa系数( $\kappa$ )。每个实验使用不同的随机种子重复10次，分类结果以平均值和标准差呈现。

## 4.3. 参数分析

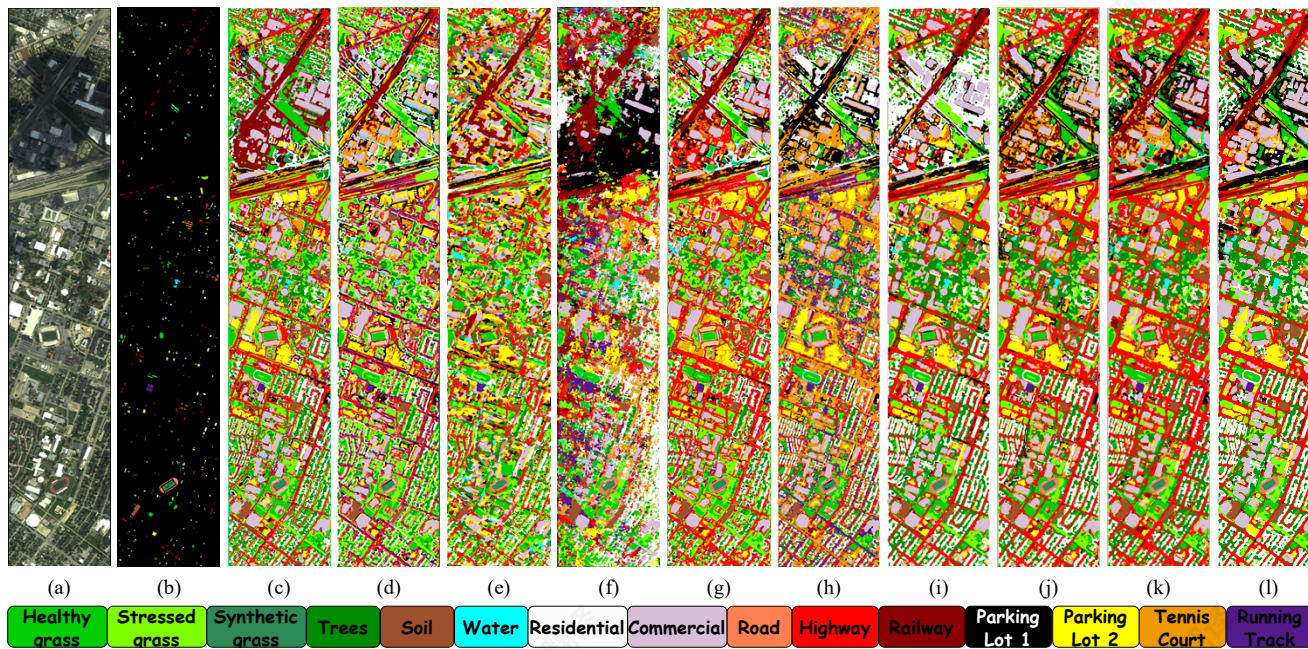
(1) **不同值  $k$  的影响。**在TSFTB模块，选择  $k$  标记的核心参数之一是识别  $k$  的最优值。为了确定  $k$  的最优值，我们



Table 4

Quantitative classification results of the Houston dataset. Best results are shown in **bold**.

| Class        | 3-DCNN     | SSFCN        | SACNet       | FcontNet     | SpectralFormer | SSFTT       | GAHT       | MorphFormer | GSCViT            | DSFormer          |
|--------------|------------|--------------|--------------|--------------|----------------|-------------|------------|-------------|-------------------|-------------------|
| 1            | 91.96±6.02 | 94.17 ± 1.94 | 87.38 ± 1.43 | 94.44 ± 1.57 | 94.32±3.37     | 87.76±5.05  | 93.61±4.89 | 96.79±1.73  | 96.81±1.95        | <b>98.25±1.26</b> |
| 2            | 92.57±6.08 | 96.80 ± 2.54 | 79.50 ± 3.08 | 90.35 ± 1.89 | 96.45±1.04     | 94.97±5.43  | 97.52±3.83 | 97.57±3.50  | 99.31±0.86        | <b>99.37±0.34</b> |
| 3            | 99.21±0.95 | 99.09 ± 0.49 | 98.56 ± 0.87 | 99.41 ± 0.64 | 99.43±0.42     | 90.36±17.37 | 99.57±0.49 | 99.32±0.90  | 99.58±0.54        | <b>99.61±0.26</b> |
| 4            | 93.58±2.12 | 89.61 ± 1.12 | 82.05 ± 3.02 | 90.59 ± 2.89 | 93.72±2.53     | 97.85±2.74  | 96.81±2.57 | 96.73±2.60  | 99.39±0.65        | <b>99.56±0.67</b> |
| 5            | 98.98±0.83 | 97.23 ± 0.58 | 82.10 ± 3.33 | 97.87 ± 0.80 | 98.70±0.92     | 98.72±1.85  | 99.66±0.57 | 99.35±0.82  | <b>100</b>        | 99.87±0.24        |
| 6            | 92.95±5.28 | 91.78 ± 1.09 | 89.20 ± 2.17 | 95.85 ± 1.68 | 96.33±1.87     | 97.09±3.11  | 98.95±1.18 | 98.87±1.19  | <b>99.96±0.11</b> | 99.78±0.65        |
| 7            | 87.83±2.82 | 82.31 ± 1.36 | 73.04 ± 1.83 | 85.48 ± 2.66 | 78.10±4.61     | 87.73±6.38  | 94.11±2.77 | 93.45±2.06  | 95.48±2.67        | <b>97.95±1.06</b> |
| 8            | 72.97±6.39 | 83.15 ± 2.30 | 62.19 ± 2.66 | 80.45 ± 3.09 | 84.86±3.65     | 78.49±5.71  | 89.36±2.98 | 87.24±4.22  | 89.66±4.75        | <b>92.28±2.94</b> |
| 9            | 83.60±4.00 | 81.41 ± 1.20 | 68.82 ± 2.52 | 80.05 ± 2.10 | 76.15±4.90     | 89.54±5.53  | 90.87±3.73 | 90.52±2.48  | 90.87±4.30        | <b>94.38±2.07</b> |
| 10           | 85.53±5.68 | 88.07 ± 1.97 | 90.93 ± 2.32 | 92.56 ± 3.15 | 89.97±3.90     | 63.59±27.82 | 93.02±5.28 | 96.29±2.30  | <b>98.09±1.25</b> | 97.69±3.11        |
| 11           | 71.20±4.31 | 84.08 ± 2.49 | 84.14 ± 1.93 | 94.36 ± 2.81 | 76.36±3.41     | 82.01±14.44 | 94.18±2.62 | 93.79±2.55  | 96.25±2.02        | <b>98.03±1.81</b> |
| 12           | 84.33±4.93 | 79.37 ± 2.80 | 84.57 ± 2.84 | 94.61 ± 1.99 | 80.35±4.81     | 67.31±29.28 | 92.53±3.96 | 94.22±3.13  | 93.44±4.63        | <b>95.16±2.97</b> |
| 13           | 94.84±0.66 | 88.76 ± 1.74 | 82.98 ± 3.61 | 97.28 ± 1.33 | 76.11±4.69     | 96.56±2.74  | 97.18±1.48 | 98.42±1.60  | 98.59±1.65        | <b>99.38±0.66</b> |
| 14           | 98.68±1.31 | 98.39 ± 0.48 | 98.92 ± 0.69 | <b>100</b>   | 96.85±1.76     | 99.79±0.39  | <b>100</b> | 99.81±0.56  | <b>100</b>        | 99.97±0.08        |
| 15           | 99.64±0.39 | 95.77 ± 0.60 | 94.23 ± 2.15 | 99.64 ± 0.32 | 98.84±1.08     | 99.82±0.21  | 99.89±0.18 | 99.69±0.42  | 99.66±0.42        | <b>100</b>        |
| OA (%)       | 88.14±1.32 | 88.89±1.00   | 81.74±0.93   | 91.47±1.32   | 88.10±0.74     | 86.72±4.09  | 94.99±0.91 | 95.36±0.99  | 96.52±0.60        | <b>97.66±0.51</b> |
| $\kappa$ (%) | 87.17±1.43 | 87.99±1.08   | 80.26±1.00   | 90.78±1.42   | 87.14±0.80     | 85.65±4.41  | 94.58±0.99 | 94.98±1.07  | 96.23±0.65        | <b>97.47±0.56</b> |
| AA (%)       | 89.86±1.18 | 90.00±0.85   | 83.91±0.90   | 92.86±1.07   | 89.10±0.68     | 88.77±3.57  | 95.82±0.81 | 96.14±0.88  | 97.14±0.49        | <b>98.09±0.42</b> |



**Figure 8:** Visualization of the classification maps on the Houston dataset. (a) False-color image. (b) Ground truth. (c) 3-DCNN. (d) SSFCN. (e) SACNet. (f) FcontNet. (g) SpectralFormer. (h) SSFTT. (i) GAHT. (j) MorphFormer. (k) GSCViT. (l) DSFormer.

performed four experiments with  $k$  set to 0.2, 0.4, 0.6, 0.8, and 1, respectively. When  $k$  is set to 0.8, the TSFTB module selects the most relevant 80% tokens for the spatial-spectral context modeling, while the remaining 20% are considered the least relevant and are therefore excluded from the information interaction process. When  $k$  is set to 1, the attention mechanism in the TSFTB module degenerates into the conventional full self-attention form. Fig. 5 presents the OA results in four datasets with different values of  $k$ . The experimental results show that when  $k$  is set to 0.2, the classification accuracy is relatively low due to the insufficient information available for long-range modeling. As the value of  $k$  increases, the OA improves, indicating that more pixels contribute to the fusion of contextual relationships, enabling the network to capture richer and more effective information

for the classification task. However, when  $k$  exceeds the optimal value, OA begins to decline, likely due to the inclusion of irrelevant or redundant information, which negatively impacts classification performance. Finally, based on the experimental results, we selected  $k$  values of 0.4, 0.6, 0.8, and 0.8 across the four datasets to achieve optimal classification performance. Furthermore, we observe that performance in most  $k$  configurations outperforms the traditional self-attention mechanism ( $k = 1$ ), validating the effectiveness of the token selection strategy.

**(2) Effect of Different Group Numbers  $g$ .** A critical parameter in the DSFormer architecture is the number of groups, denoted as  $g$ . To determine the optimal value for  $g$ , we conducted four experimental configurations with  $g$  set to 1, 2, 4, and 8, respectively. By utilizing grouping and

表格4

休斯顿数据集的定量分类结果。最佳结果以粗体显示。

| 类别           | 3-DCNN     | SSFCN        | SACNet       | FcontNet     | SpectralFormer | SSFTT       | GAHT       | MorphFormer | GSCViT            | DSFormer          |
|--------------|------------|--------------|--------------|--------------|----------------|-------------|------------|-------------|-------------------|-------------------|
| 1            | 91.96±6.02 | 94.17 ± 1.94 | 87.38 ± 1.43 | 94.44 ± 1.57 | 94.32±3.37     | 87.76±5.05  | 93.61±4.89 | 96.79±1.73  | 96.81±1.95        | <b>98.25±1.26</b> |
| 2            | 92.57±6.08 | 96.80 ± 2.54 | 79.50 ± 3.08 | 90.35 ± 1.89 | 96.45±1.04     | 94.97±5.43  | 97.52±3.83 | 97.57±3.50  | 99.31±0.86        | <b>99.37±0.34</b> |
| 3            | 99.21±0.95 | 99.09 ± 0.49 | 98.56 ± 0.87 | 99.41 ± 0.64 | 99.43±0.42     | 90.36±17.37 | 99.57±0.49 | 99.32±0.90  | 99.58±0.54        | <b>99.61±0.26</b> |
| 4            | 93.58±2.12 | 89.61 ± 1.12 | 82.05 ± 3.02 | 90.59 ± 2.89 | 93.72±2.53     | 97.85±2.74  | 96.81±2.57 | 96.73±2.60  | 99.39±0.65        | <b>99.56±0.67</b> |
| 5            | 98.98±0.83 | 97.23 ± 0.58 | 82.10 ± 3.33 | 97.87 ± 0.80 | 98.70±0.92     | 98.72±1.85  | 99.66±0.57 | 99.35±0.82  | <b>100</b>        | 99.87±0.24        |
| 6            | 92.95±5.28 | 91.78 ± 1.09 | 89.20 ± 2.17 | 95.85 ± 1.68 | 96.33±1.87     | 97.09±3.11  | 98.95±1.18 | 98.87±1.19  | <b>99.96±0.11</b> | 99.78±0.65        |
| 7            | 87.83±2.82 | 82.31 ± 1.36 | 73.04 ± 1.83 | 85.48 ± 2.66 | 78.10±4.61     | 87.73±6.38  | 94.11±2.77 | 93.45±2.06  | 95.48±2.67        | <b>97.95±1.06</b> |
| 8            | 72.97±6.39 | 83.15 ± 2.30 | 62.19 ± 2.66 | 80.45 ± 3.09 | 84.86±3.65     | 78.49±5.71  | 89.36±2.98 | 87.24±4.22  | 89.66±4.75        | <b>92.28±2.94</b> |
| 9            | 83.60±4.00 | 81.41 ± 1.20 | 68.82 ± 2.52 | 80.05 ± 2.10 | 76.15±4.90     | 89.54±5.53  | 90.87±3.73 | 90.52±2.48  | 90.87±4.30        | <b>94.38±2.07</b> |
| 10           | 85.53±5.68 | 88.07 ± 1.97 | 90.93 ± 2.32 | 92.56 ± 3.15 | 89.97±3.90     | 63.59±27.82 | 93.02±5.28 | 96.29±2.30  | <b>98.09±1.25</b> | 97.69±3.11        |
| 11           | 71.20±4.31 | 84.08 ± 2.49 | 84.14 ± 1.93 | 94.36 ± 2.81 | 76.36±3.41     | 82.01±14.44 | 94.18±2.62 | 93.79±2.55  | 96.25±2.02        | <b>98.03±1.81</b> |
| 12           | 84.33±4.93 | 79.37 ± 2.80 | 84.57 ± 2.84 | 94.61 ± 1.99 | 80.35±4.81     | 67.31±29.28 | 92.53±3.96 | 94.22±3.13  | 93.44±4.63        | <b>95.16±2.97</b> |
| 13           | 94.84±0.66 | 88.76 ± 1.74 | 82.98 ± 3.61 | 97.28 ± 1.33 | 76.11±4.69     | 96.56±2.74  | 97.18±1.48 | 98.42±1.60  | 98.59±1.65        | <b>99.38±0.66</b> |
| 14           | 98.68±1.31 | 98.39 ± 0.48 | 98.92 ± 0.69 | <b>100</b>   | 96.85±1.76     | 99.79±0.39  | <b>100</b> | 99.81±0.56  | <b>100</b>        | 99.97±0.08        |
| 15           | 99.64±0.39 | 95.77 ± 0.60 | 94.23 ± 2.15 | 99.64 ± 0.32 | 98.84±1.08     | 99.82±0.21  | 99.89±0.18 | 99.69±0.42  | 99.66±0.42        | <b>100</b>        |
| OA (%)       | 88.14±1.32 | 88.89±1.00   | 81.74±0.93   | 91.47±1.32   | 88.10±0.74     | 86.72±4.09  | 94.99±0.91 | 95.36±0.99  | 96.52±0.60        | <b>97.66±0.51</b> |
| $\kappa$ (%) | 87.17±1.43 | 87.99±1.08   | 80.26±1.00   | 90.78±1.42   | 87.14±0.80     | 85.65±4.41  | 94.58±0.99 | 94.98±1.07  | 96.23±0.65        | <b>97.47±0.56</b> |
| AA (%)       | 89.86±1.18 | 90.00±0.85   | 83.91±0.90   | 92.86±1.07   | 89.10±0.68     | 88.77±3.57  | 95.82±0.81 | 96.14±0.88  | 97.14±0.49        | <b>98.09±0.42</b> |

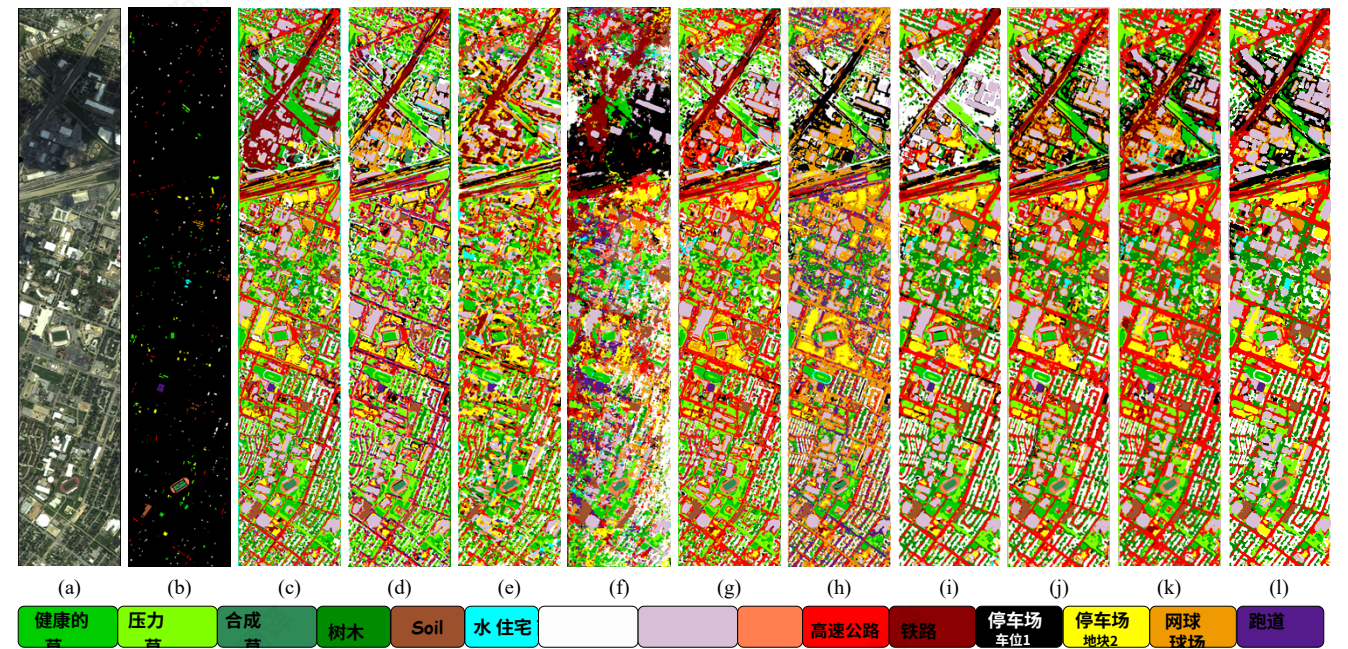


图8：在休斯顿数据集上对分类图进行可视化。(a) 假彩色图像。(b) 真实情况。(c) 3-DCNN。(d) SSFCN。(e) SACNet。(f) FcontNet。(g) SpectralFormer。(h) SSFTT。(i) GAHT。(j) MorphFormer。(k) GSCViT。(l) DSFormer。

分别以  $k$  设置为0.2、0.4、0.6、0.8和1进行了四个实验。当  $k$  设置为0.8时，TSFTB模块为空间-频谱上下文建模选择最相关的80%标记，而其余20%被认为最不相关，因此从信息交互过程中被排除。当  $k$  设置为1时，TSFTB模块中的注意力机制退化为传统的全自注意力形式。图5展示了四个不同  $k$  值的数据集上的OA结果。实验结果表明，当  $k$  设置为0.2时，由于长距离建模可用信息不足，分类准确率相对较低。随着  $k$  值的增加，OA得到提升，表明更多像素有助于上下文关系的融合，使网络能够捕获更丰富和有效的信息

用于分类任务。然而，当  $k$  超过最优值时，OA 开始下降，可能是由于包含了无关或冗余信息，这会降低分类性能。最后，根据实验结果，我们在四个数据集上选择了  $k$  值为 0.4、0.6、0.8 和 0.8，以实现最优分类性能。此外，我们观察到大多数  $k$  配置的性能优于传统的自注意力机制 ( $k = 1$ )，验证了词选择策略的有效性。

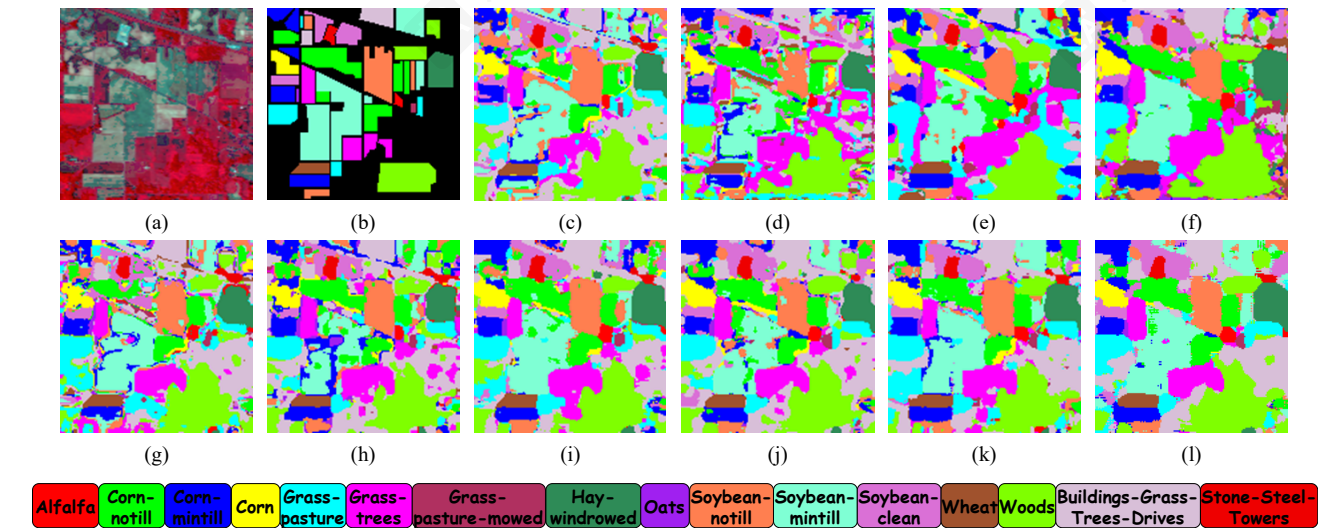
**(2) 不同组数  $g$  的影响。** DSFormer 架构中的一个关键参数是组数，记作  $g$ 。为了确定  $g$  的最优值，我们进行了四种实验配置，将  $g$  分别设置为 1、2、4 和 8。通过利用分组和



**Table 5**

Quantitative classification results of the Indian Pines dataset. Best results are shown in **Bold**.

| Class        | 3-DCNN     | SSFCN        | SACNet       | FcontNet     | SpectralFormer | SSFTT       | GAHT       | MorphFormer       | GSCViT            | DSFormer          |
|--------------|------------|--------------|--------------|--------------|----------------|-------------|------------|-------------------|-------------------|-------------------|
| 1            | 93.87±2.26 | 93.23 ± 2.26 | 99.68 ± 0.97 | 90.32 ± 6.12 | 90.00±7.83     | 96.77±4.33  | <b>100</b> | 97.10±3.04        | 99.03±1.48        | 99.35±1.29        |
| 2            | 60.45±9.78 | 75.64 ± 1.76 | 89.33 ± 1.66 | 80.86 ± 6.15 | 68.72±3.71     | 84.45±7.24  | 88.08±3.69 | 88.15±2.65        | 89.47±2.85        | <b>93.22±2.14</b> |
| 3            | 70.64±4.44 | 83.36 ± 2.00 | 93.59 ± 1.95 | 94.21 ± 2.93 | 82.14±6.69     | 79.05±26.67 | 93.76±3.85 | 94.05±5.19        | <b>96.53±2.67</b> | 96.08±2.40        |
| 4            | 95.72±3.80 | 92.51 ± 3.08 | 97.86 ± 1.29 | 95.29 ± 1.24 | 94.60±2.40     | 97.65±2.74  | 99.63±0.79 | <b>99.89±0.32</b> | <b>99.89±0.21</b> | 99.79±0.43        |
| 5            | 89.49±2.49 | 90.30 ± 3.39 | 97.76 ± 0.75 | 94.83 ± 3.19 | 90.81±2.73     | 95.38±2.33  | 96.56±2.07 | 96.95±1.59        | 97.23±1.69        | <b>98.48±1.45</b> |
| 6            | 96.85±0.64 | 89.75 ± 1.65 | 98.65 ± 0.79 | 96.74 ± 2.12 | 95.32±1.80     | 96.82±5.89  | 99.34±0.42 | 98.63±1.28        | 99.43±0.61        | <b>99.68±0.22</b> |
| 7            | 96.92±3.77 | 98.46 ± 3.08 | <b>100</b>   | <b>100</b>   | 98.46±4.62     | <b>100</b>  | <b>100</b> | <b>100</b>        | <b>100</b>        | <b>100</b>        |
| 8            | 98.06±1.90 | 94.35 ± 1.40 | 98.62 ± 0.82 | 99.77 ± 0.26 | 97.27±2.36     | 99.58±0.55  | 99.86±0.28 | 99.74±0.29        | 99.93±0.21        | <b>100</b>        |
| 9            | <b>100</b> | <b>100</b>   | <b>100</b>   | <b>100</b>   | 98.00±6.00     | <b>100</b>  | <b>100</b> | <b>100</b>        | <b>100</b>        | <b>100</b>        |
| 10           | 76.80±6.72 | 75.42 ± 2.65 | 89.34 ± 2.62 | 93.67 ± 2.42 | 81.40±3.86     | 88.37±5.57  | 91.91±2.68 | 91.34±3.36        | <b>94.03±2.51</b> | <b>94.03±2.55</b> |
| 11           | 66.88±7.56 | 64.51 ± 3.09 | 91.06 ± 1.24 | 89.84 ± 5.82 | 73.53±4.42     | 60.80±25.84 | 82.13±4.48 | 87.42±4.85        | 89.50±3.21        | <b>90.25±1.53</b> |
| 12           | 82.71±7.42 | 80.41 ± 5.46 | 86.10 ± 1.31 | 91.20 ± 2.78 | 74.84±6.65     | 91.14±3.34  | 94.97±2.23 | 95.49±1.51        | 95.82±1.88        | <b>96.24±1.73</b> |
| 13           | 99.87±0.26 | 98.39 ± 1.09 | 99.61 ± 0.43 | <b>100</b>   | 99.42±0.54     | 99.94±0.19  | <b>100</b> | 99.94±0.19        | 99.94±0.19        | <b>100</b>        |
| 14           | 92.24±2.85 | 89.34 ± 1.18 | 96.02 ± 1.04 | 95.74 ± 1.77 | 92.10±2.82     | 96.29±2.00  | 95.62±1.83 | 97.42±1.54        | 97.77±0.88        | <b>98.52±0.66</b> |
| 15           | 94.85±4.19 | 90.65 ± 2.18 | 99.97 ± 0.09 | 99.52 ± 0.40 | 93.45±4.24     | 97.11±1.79  | 98.72±1.20 | 99.38±0.80        | <b>99.88±0.15</b> | <b>99.88±0.15</b> |
| 16           | <b>100</b> | 99.30 ± 1.07 | <b>100</b>   | 99.07 ± 1.14 | 99.53±0.93     | 99.77±0.70  | <b>100</b> | 99.77±0.70        | <b>100</b>        | <b>100</b>        |
| OA (%)       | 78.28±2.01 | 79.40±1.30   | 93.05±0.60   | 91.93±3.22   | 81.88±1.36     | 83.47±9.07  | 91.39±1.32 | 92.91±1.52        | 94.26±0.89        | <b>95.17±0.93</b> |
| $\kappa$ (%) | 75.30±2.26 | 76.68±1.45   | 92.04±0.68   | 90.78±3.66   | 79.43±1.51     | 81.43±9.94  | 90.17±1.48 | 91.89±1.73        | 93.43±1.01        | <b>94.47±1.07</b> |
| AA (%)       | 88.46±1.28 | 88.48±1.09   | 96.10±0.35   | 95.06±1.91   | 89.35±1.10     | 92.70±3.71  | 96.29±0.45 | 96.54±0.82        | 97.37±0.39        | <b>97.85±0.44</b> |



**Figure 9:** Visualization of the classification maps on the Indian Pines dataset. (a) False-color image. (b) Ground truth. (c) 3-DCNN. (d) SSFCN. (e) SACNet. (f) FcontNet. (g) SpectralFormer. (h) SSFTT. (i) GAHT. (j) MorphFormer. (k) GSCViT. (l) DSFormer.

3D convolutions, the TSFTB module preserves the spatial-spectral characteristics of HSIs, enabling the extraction of tokens that incorporate both spatial and spectral features. The value of  $g$  directly impacts the size of the attention matrix, with larger  $g$  values leading to increased computational complexity due to the expansion of the matrix. When  $g$  is set to 1, the TSFTB module reduces to purely spatial token selection without any grouping. The experimental results, as illustrated in Fig. 6, show that when  $g$  is set to 1, no grouping strategy is applied to preserve spatial-spectral characteristics, resulting in a relatively lower classification accuracy across the four datasets. In contrast, the grouping strategy significantly improves classification performance. Notably, the optimal classification performance is achieved with  $g$  is set to 4 across all four datasets, while maintaining relatively efficient computational complexity.

#### 4.4. Ablation Study

**(1) Performance Contribution of Each Module.** We explore the impact of each module within DSFormer on classification performance through an ablation study, as shown in Table 1. The baseline architecture utilizes a single  $3 \times 3$  convolutional layer, as well as a variant that employs nine consecutively stacked  $3 \times 3$  convolutional layers to align with the depth of the DSFormer network. The results presented indicate that both the KSFTB and TSFTB modules significantly enhance the network's classification performance. For example, on the Pavia University dataset, the baseline with single convolutional layer achieves an OA of only 87.67%. However, increasing the number of network layers did not lead to improved accuracy. In fact, employing nine convolutional layers yielded an OA of only 86.33%. The introduction of the KSFTB module improved the OA to 93.44%, while the application of the TSFTB module increased the OA to 95.81%. Each of these modules contributes to improved

**表 5**

印度针叶树数据集的定量分类结果。最佳结果以粗体显示。

| 类别           | 3-DCNN     | SSFCN        | SACNet       | FcontNet     | SpectralFormer | SSFTT       | GAHT       | MorphFormer       | GSCViT            | DSFormer          |
|--------------|------------|--------------|--------------|--------------|----------------|-------------|------------|-------------------|-------------------|-------------------|
| 1            | 93.87±2.26 | 93.23 ± 2.26 | 99.68 ± 0.97 | 90.32 ± 6.12 | 90.00±7.83     | 96.77±4.33  | <b>100</b> | 97.10±3.04        | 99.03±1.48        | 99.35±1.29        |
| 2            | 60.45±9.78 | 75.64 ± 1.76 | 89.33 ± 1.66 | 80.86 ± 6.15 | 68.72±3.71     | 84.45±7.24  | 88.08±3.69 | 88.15±2.65        | 89.47±2.85        | <b>93.22±2.14</b> |
| 3            | 70.64±4.44 | 83.36 ± 2.00 | 93.59 ± 1.95 | 94.21 ± 2.93 | 82.14±6.69     | 79.05±26.67 | 93.76±3.85 | 94.05±5.19        | <b>96.53±2.67</b> | 96.08±2.40        |
| 4            | 95.72±3.80 | 92.51 ± 3.08 | 97.86 ± 1.29 | 95.29 ± 1.24 | 94.60±2.40     | 97.65±2.74  | 99.63±0.79 | <b>99.89±0.32</b> | <b>99.89±0.21</b> | 99.79±0.43        |
| 5            | 89.49±2.49 | 90.30 ± 3.39 | 97.76 ± 0.75 | 94.83 ± 3.19 | 90.81±2.73     | 95.38±2.33  | 96.56±2.07 | 96.95±1.59        | 97.23±1.69        | <b>98.48±1.45</b> |
| 6            | 96.85±0.64 | 89.75 ± 1.65 | 98.65 ± 0.79 | 96.74 ± 2.12 | 95.32±1.80     | 96.82±5.89  | 99.34±0.42 | 98.63±1.28        | 99.43±0.61        | <b>99.68±0.22</b> |
| 7            | 96.92±3.77 | 98.46 ± 3.08 | <b>100</b>   | <b>100</b>   | 98.46±4.62     | <b>100</b>  | <b>100</b> | <b>100</b>        | <b>100</b>        | <b>100</b>        |
| 8            | 98.06±1.90 | 94.35 ± 1.40 | 98.62 ± 0.82 | 99.77 ± 0.26 | 97.27±2.36     | 99.58±0.55  | 99.86±0.28 | 99.74±0.29        | 99.93±0.21        | <b>100</b>        |
| 9            | <b>100</b> | <b>100</b>   | <b>100</b>   | <b>100</b>   | 98.00±6.00     | <b>100</b>  | <b>100</b> | <b>100</b>        | <b>100</b>        | <b>100</b>        |
| 10           | 76.80±6.72 | 75.42 ± 2.65 | 89.34 ± 2.62 | 93.67 ± 2.42 | 81.40±3.86     | 88.37±5.57  | 91.91±2.68 | 91.34±3.36        | <b>94.03±2.51</b> | <b>94.03±2.55</b> |
| 11           | 66.88±7.56 | 64.51 ± 3.09 | 91.06 ± 1.24 | 89.84 ± 5.82 | 73.53±4.42     | 60.80±25.84 | 82.13±4.48 | 87.42±4.85        | 89.50±3.21        | <b>90.25±1.53</b> |
| 12           | 82.71±7.42 | 80.41 ± 5.46 | 86.10 ± 1.31 | 91.20 ± 2.78 | 74.84±6.65     | 91.14±3.34  | 94.97±2.23 | 95.49±1.51        | 95.82±1.88        | <b>96.24±1.73</b> |
| 13           | 99.87±0.26 | 98.39 ± 1.09 | 99.61 ± 0.43 | <b>100</b>   | 99.42±0.54     | 99.94±0.19  | <b>100</b> | 99.94±0.19        | 99.94±0.19        | <b>100</b>        |
| 14           | 92.24±2.85 | 89.34 ± 1.18 | 96.02 ± 1.04 | 95.74 ± 1.77 | 92.10±2.82     | 96.29±2.00  | 95.62±1.83 | 97.42±1.54        | 97.77±0.88        | <b>98.52±0.66</b> |
| 15           | 94.85±4.19 | 90.65 ± 2.18 | 99.97 ± 0.09 | 99.52 ± 0.40 | 93.45±4.24     | 97.11±1.79  | 98.72±1.20 | 99.38±0.80        | <b>99.88±0.15</b> | <b>99.88±0.15</b> |
| 16           | <b>100</b> | 99.30 ± 1.07 | <b>100</b>   | 99.07 ± 1.14 | 99.53±0.93     | 99.77±0.70  | <b>100</b> | 99.77±0.70        | <b>100</b>        | <b>100</b>        |
| OA (%)       | 78.28±2.01 | 79.40±1.30   | 93.05±0.60   | 91.93±3.22   | 81.88±1.36     | 83.47±9.07  | 91.39±1.32 | 92.91±1.52        | 94.26±0.89        | <b>95.17±0.93</b> |
| $\kappa$ (%) | 75.30±2.26 | 76.68±1.45   | 92.04±0.68   | 90.78±3.66   | 79.43±1.51     | 81.43±9.94  | 90.17±1.48 | 91.89±1.73        | 93.43±1.01        | <b>94.47±1.07</b> |
| AA (%)       | 88.46±1.28 | 88.48±1.09   | 96.10±0.35   | 95.06±1.91   | 89.35±1.10     | 92.70±3.71  | 96.29±0.45 | 96.54±0.82        | 97.37±0.39        | <b>97.85±0.44</b> |

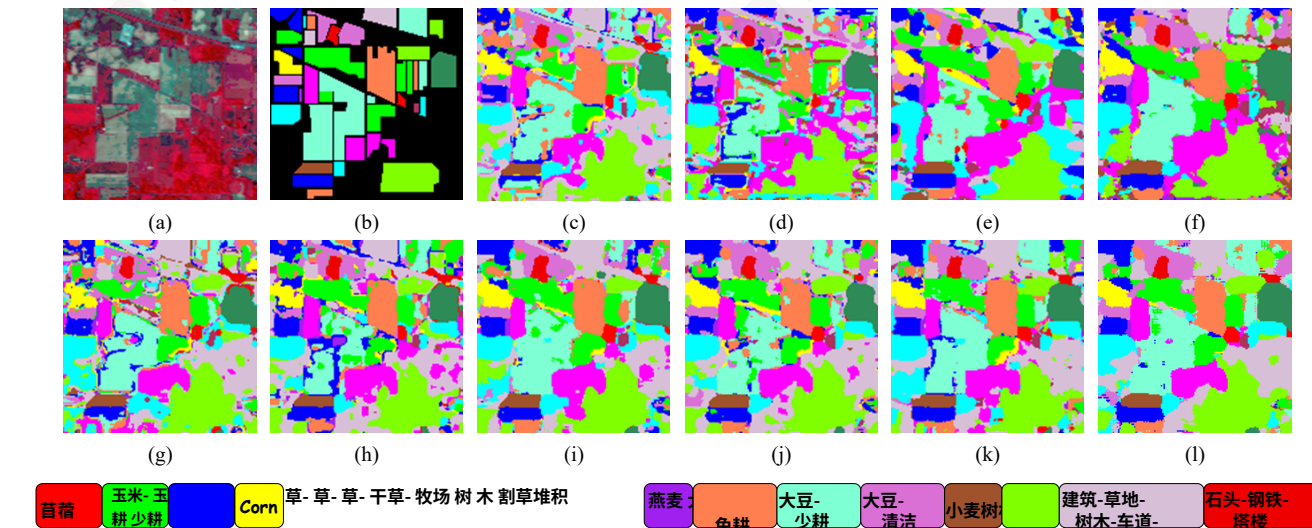


图9：在印度针叶树数据集上的分类图可视化。(a) 假彩色图像。(b) 真实情况。(c) 3-DCNN。(d) SSFCN。(e) SACNet。(f) FcontNet。(g) SpectralFormer。(h) SSFTT。(i) GAHT。(j) MorphFormer。(k) GSCViT。(l) DSFormer。

3D卷积，TSFTB 模块保留了高光谱图像的时空频特征，能够提取包含空间和光谱特征的标记。 $g$  的值直接影响注意力矩阵的大小，较大的  $g$  值会导致矩阵扩展，从而增加计算复杂度。当  $g$  设置为 1 时，TSFTB 模块简化为纯粹的空间标记选择，没有任何分组。实验结果如图 6 所示，当  $g$  设置为 1 时，没有应用分组策略以保留时空频特征，导致在四个数据集上的分类精度相对较低。相比之下，分组策略显著提高了分类性能。值得注意的是，在所有四个数据集上，当  $g$  设置为 4 时，实现了最优的分类性能，同时保持了相对高效的计算复杂度。

#### 4.4. 消融实验

**(1) 各模块的性能贡献。**我们通过消融实验探索了 DSFormer 中每个模块对分类性能的影响，如表 1 所示。基线架构使用单个  $3 \times 3$  卷积层，以及一个采用九个连续堆叠的  $3 \times 3$  卷积层的变体，与 DSFormer 网络的深度保持一致。结果表明，KSFTB 和 TSFTB 模块都显著提升了网络的分类性能。例如，在帕维亚大学数据集上，使用单个卷积层的基线模型仅达到 OA 87.67%。然而，增加网络层数并未提高准确率。事实上，使用九个卷积层得到的 OA 仅为 86.33%。引入 KSFTB 模块将 OA 提升至 93.44%，而应用 TSFTB 模块则将 OA 提高到 95.81%。这些模块中的每一个都对改进



**Table 6**  
Quantitative classification results of the Whu-HongHu dataset. Best results are shown in **bold**.

| Class        | 3-DCNN     | SSFCN        | SACNet       | FcontNet            | SpectralFormer | SSFTT       | GAHT        | MorphFormer  | GSCViT            | DSFormer          |
|--------------|------------|--------------|--------------|---------------------|----------------|-------------|-------------|--------------|-------------------|-------------------|
| 1            | 90.78±2.65 | 92.34 ± 1.76 | 88.22 ± 1.86 | 89.94 ± 2.38        | 95.28±1.04     | 82.37±15.01 | 96.13±0.81  | 94.97±4.13   | 93.46±5.86        | <b>96.35±0.77</b> |
| 2            | 92.10±1.50 | 88.75 ± 1.34 | 89.69 ± 1.26 | 88.54 ± 5.40        | 90.01±2.12     | 69.44±31.34 | 96.23±1.47  | 95.61±2.81   | 95.07±4.31        | <b>96.84±1.23</b> |
| 3            | 80.27±7.44 | 81.16 ± 0.94 | 85.64 ± 1.10 | 85.93 ± 0.73        | 78.32±1.68     | 80.07±11.96 | 84.00±8.94  | 87.78±4.15   | 89.40±2.69        | <b>92.66±1.20</b> |
| 4            | 90.10±2.58 | 87.93 ± 1.28 | 91.50 ± 1.90 | 89.94 ± 5.18        | 93.20±2.40     | 84.41±11.17 | 95.46±0.70  | 93.31±2.80   | 95.56±5.62        | <b>97.09±1.01</b> |
| 5            | 85.49±3.52 | 84.99 ± 1.04 | 90.95 ± 0.75 | 93.46 ± 5.21        | 85.74±5.17     | 72.78±19.67 | 93.45±2.38  | 94.44±3.14   | 90.68±9.77        | <b>96.76±1.79</b> |
| 6            | 90.33±4.36 | 85.74 ± 1.27 | 86.64 ± 1.04 | 91.05 ± 2.85        | 90.41±2.92     | 66.43±29.85 | 94.12±2.52  | 94.68±2.32   | 93.32±2.90        | <b>94.95±1.93</b> |
| 7            | 66.40±3.44 | 61.18 ± 2.27 | 57.62 ± 2.61 | 76.95 ± 3.22        | 70.49±3.32     | 69.35±14.28 | 78.60±3.92  | 78.57±3.69   | 80.01±5.05        | <b>84.44±1.99</b> |
| 8            | 59.23±6.12 | 51.81 ± 2.87 | 81.22 ± 3.34 | 97.54 ± 1.79        | 63.43±3.11     | 64.45±12.07 | 80.02±4.75  | 84.93±7.33   | 83.12±3.24        | <b>92.69±2.15</b> |
| 9            | 95.45±1.16 | 95.04 ± 0.74 | 87.93 ± 1.76 | 92.61 ± 1.73        | 96.94±0.86     | 95.58±3.40  | 97.27±1.74  | 96.12±2.53   | <b>97.55±1.37</b> | 97.21±1.80        |
| 10           | 82.54±4.28 | 66.45 ± 2.86 | 59.82 ± 2.38 | 86.82 ± 2.46        | 61.54±5.74     | 81.11±8.85  | 89.57±2.24  | 88.08±5.46   | 82.34±11.69       | <b>93.54±1.42</b> |
| 11           | 68.36±5.18 | 56.68 ± 2.51 | 57.55 ± 3.51 | 80.69 ± 4.14        | 71.51±3.36     | 55.64±19.38 | 85.85±3.57  | 87.15±4.04   | 87.65±3.80        | <b>92.58±1.61</b> |
| 12           | 70.88±5.60 | 58.97 ± 2.42 | 78.58 ± 3.28 | 89.90 ± 3.60        | 70.96±4.17     | 56.33±20.32 | 79.35±5.21  | 82.53±5.54   | 81.50±9.10        | <b>88.52±2.96</b> |
| 13           | 66.08±7.33 | 54.73 ± 1.88 | 67.93 ± 2.28 | 86.36 ± 4.36        | 67.83±5.20     | 67.83±5.20  | 78.49±4.74  | 80.09±4.50   | 78.77±4.33        | <b>84.46±2.80</b> |
| 14           | 86.10±5.35 | 76.66 ± 2.60 | 78.39 ± 1.78 | 95.91 ± 2.47        | 85.54±3.17     | 85.43±8.74  | 94.55±2.26  | 95.28±3.66   | 92.92±5.11        | <b>97.54±0.91</b> |
| 15           | 94.50±1.95 | 93.32 ± 1.00 | 98.58 ± 0.83 | 99.35 ± 0.47        | 94.02±2.90     | 97.13±2.83  | 98.26±1.01  | 99.35 ± 0.47 | 94.02±2.90        | <b>99.26±0.76</b> |
| 16           | 91.36±3.75 | 89.41 ± 1.53 | 90.74 ± 2.92 | 94.29 ± 2.02        | 89.36±3.51     | 91.33±3.20  | 94.82±1.69  | 94.05±4.07   | 97.26±1.37        | <b>98.31±1.48</b> |
| 17           | 91.41±3.83 | 89.81 ± 2.16 | 95.75 ± 1.78 | 95.14 ± 2.61        | 89.21±2.37     | 92.08±3.87  | 96.65±1.96  | 97.54±1.77   | 95.30±5.49        | <b>97.66±1.49</b> |
| 18           | 91.13±2.78 | 90.33 ± 2.19 | 94.03 ± 0.81 | 97.23 ± 1.67        | 90.66±3.82     | 64.21±25.62 | 95.68±2.52  | 97.23 ± 1.67 | 90.66±3.82        | <b>98.80±0.90</b> |
| 19           | 89.90±2.60 | 85.11 ± 1.81 | 81.46 ± 1.40 | 86.18 ± 2.97        | 90.03±1.94     | 67.38±24.16 | 94.89±0.94  | 94.36±1.48   | 94.21±2.12        | <b>95.80±0.74</b> |
| 20           | 92.40±2.76 | 96.38 ± 0.78 | 91.49 ± 2.31 | <b>99.56 ± 0.16</b> | 94.45±2.00     | 75.31±22.05 | 95.69±2.37  | 96.27±2.90   | 97.42±1.69        | 98.90±0.52        |
| 21           | 83.04±9.61 | 87.71 ± 1.64 | 95.23 ± 2.22 | <b>99.69 ± 0.20</b> | 87.86±5.67     | 55.38±36.65 | 93.64±12.16 | 92.61±15.09  | 89.48±11.80       | 99.01±0.66        |
| 22           | 90.53±4.40 | 87.92 ± 1.82 | 94.77 ± 3.57 | <b>99.13 ± 0.66</b> | 91.54±3.50     | 88.84±9.76  | 95.71±2.00  | 97.01±2.54   | 97.95±1.12        |                   |
| OA (%)       | 85.16±1.18 | 81.26±0.70   | 84.12±1.13   | 89.10±2.83          | 86.36±1.05     | 77.17±9.41  | 91.65±0.85  | 91.19±1.52   | 91.79±2.32        | <b>94.59±0.50</b> |
| $\kappa$ (%) | 81.67±1.39 | 76.98±0.81   | 80.33±1.31   | 86.55±3.32          | 83.03±1.24     | 72.46±10.69 | 89.55±1.05  | 89.04±1.82   | 89.76±2.72        | <b>93.20±0.62</b> |
| AA (%)       | 84.02±1.09 | 80.11±0.72   | 83.81±0.68   | 91.65±1.31          | 84.01±0.83     | 75.03±9.24  | 91.29±0.90  | 91.93±1.60   | 91.28±1.23        | <b>95.01±0.21</b> |

classification performance through their fusion mechanisms. Specifically, the KSFTB module improves recognition accuracy for objects of varying scales by enabling the network to adaptively select and fuse appropriate receptive field features from spatial and spectral respectively. While the TSFTB module enhances classification by computing self-attention spatial spectral contexts on the most relevant tokens. This process reduces redundancy and interference from irrelevant information, thus improving the integration and utilization of useful features. When both KSFTB and TSFTB modules are applied together, the classification performance is further improved, with the OA yielding 96.59%. In summary, the results of the ablation study have demonstrated that KSFTB and TSFTB modules are able to simultaneously improve the network’s classification performance on all four HSI datasets, while the combination of two modules exhibits a synergistic effect, achieving the best OAs and confirming the effectiveness of these modules in enhancing network classification performance.

**(2) Performance Contribution of Spatial-Spectral Mechanism.** We further explored the impact of the spatial-spectral selection mechanism within KSFTB on classification performance. As shown in Table 2, classification performance decreases in the four datasets when using the single spatial or spectral selection branch individually. This highlights the importance of considering both spatial and spectral information for the accurate identification of ground objects in HSIs. It is essential to not only adaptively optimal the spatial receptive field but also integrate the spectral features in appropriate scales, as the required spatial-spectral contextual information varies significantly across different land cover types. For example, in the Pavia University dataset, the OA reaches 96.09% and 96.01% respectively

when using the single spatial or spectral branches. In contrast, when both spatial and spectral receptive field fusion mechanisms are applied simultaneously, the OA reaches 96.59%. In summary, this ablation study demonstrates that optimal classification performance is achieved when both spatial and spectral multiscale features are employed, thus validating the effectiveness of the proposed KSFTB.

#### 4.5. Performance Comparison and Analysis

In this subsection, we compare the proposed method with other classical and state-of-the-art approaches. A brief introduction to each method included in the experiments is provided below.

- 1) 3D-CNN (Ben Hamida et al., 2018): Spatial-spectral classification using 3D-CNN. We follow the implementation with the original article.
- 2) SSFCN (Xu et al., 2020): Spatial-spectral classification with an end-to-end fully convolutional network. The model is trained using the Adam optimizer with a learning rate of  $5e - 4$  and a momentum of 0.9.
- 3) SACNet (Xu et al., 2021): Spatial-spectral classification method that integrates self-attention learning with context encoding. The network is trained using the Adam optimizer with a learning rate of  $5e - 4$  and a weight decay of  $5e - 5$ .
- 4) FcontNet (Wang et al., 2022a): Multiscale classification based on non-local contextual attention across space, bands, and scales. The base learning rate is 0.01, adjusted using the poly scheduling policy. The model is optimized using SGD with a momentum of 0.9 and a weight decay of  $1e - 4$ .

**表6**  
Whu-HongHu数据集的定量分类结果。最佳结果以粗体显示。

| 类别           | 3-DCNN     | SSFCN        | SACNet       | FcontNet            | SpectralFormer | SSFTT       | GAHT        | MorphFormer  | GSCViT            | DSFormer          |
|--------------|------------|--------------|--------------|---------------------|----------------|-------------|-------------|--------------|-------------------|-------------------|
| 1            | 90.78±2.65 | 92.34 ± 1.76 | 88.22 ± 1.86 | 89.94 ± 2.38        | 95.28±1.04     | 82.37±15.01 | 96.13±0.81  | 94.97±4.13   | 93.46±5.86        | <b>96.35±0.77</b> |
| 2            | 92.10±1.50 | 88.75 ± 1.34 | 89.69 ± 1.26 | 88.54 ± 5.40        | 90.01±2.12     | 69.44±31.34 | 96.23±1.47  | 95.61±2.81   | 95.07±4.31        | <b>96.84±1.23</b> |
| 3            | 80.27±7.44 | 81.16 ± 0.94 | 85.64 ± 1.10 | 85.93 ± 0.73        | 78.32±1.68     | 80.07±11.96 | 84.00±8.94  | 87.78±4.15   | 89.40±2.69        | <b>92.66±1.20</b> |
| 4            | 90.10±2.58 | 87.93 ± 1.28 | 91.50 ± 1.90 | 89.94 ± 5.18        | 93.20±2.40     | 84.41±11.17 | 95.46±0.70  | 93.31±2.80   | 95.56±5.62        | <b>97.09±1.01</b> |
| 5            | 85.49±3.52 | 84.99 ± 1.04 | 90.95 ± 0.75 | 93.46 ± 5.21        | 85.74±5.17     | 72.78±19.67 | 93.45±2.38  | 94.44±3.14   | 90.68±9.77        | <b>96.76±1.79</b> |
| 6            | 90.33±4.36 | 85.74 ± 1.27 | 86.64 ± 1.04 | 91.05 ± 2.85        | 90.41±2.92     | 66.43±29.85 | 94.12±2.52  | 94.68±2.32   | 93.32±2.90        | <b>94.95±1.93</b> |
| 7            | 66.40±3.44 | 61.18 ± 2.27 | 57.62 ± 2.61 | 76.95 ± 3.22        | 70.49±3.32     | 69.35±14.28 | 78.60±3.92  | 78.57±3.69   | 80.01±5.05        | <b>84.44±1.99</b> |
| 8            | 59.23±6.12 | 51.81 ± 2.87 | 81.22 ± 3.34 | 97.54 ± 1.79        | 63.43±3.11     | 64.45±12.07 | 80.02±4.75  | 84.93±7.33   | 83.12±3.24        | <b>92.69±2.15</b> |
| 9            | 95.45±1.16 | 95.04 ± 0.74 | 87.93 ± 1.76 | 92.61 ± 1.73        | 96.94±0.86     | 95.58±3.40  | 97.27±1.74  | 96.12±2.53   | <b>97.55±1.37</b> | 97.21±1.80        |
| 10           | 82.54±4.28 | 66.45 ± 2.86 | 59.82 ± 2.38 | 86.82 ± 2.46        | 61.54±5.74     | 81.11±8.85  | 89.57±2.24  | 88.08±5.46   | 82.34±11.69       | <b>93.54±1.42</b> |
| 11           | 68.36±5.18 | 56.68 ± 2.51 | 57.55 ± 3.51 | 80.69 ± 4.14        | 71.51±3.36     | 55.64±19.38 | 85.85±3.57  | 87.15±4.04   | 87.65±3.80        | <b>92.58±1.61</b> |
| 12           | 70.88±5.60 | 58.97 ± 2.42 | 78.58 ± 3.28 | 89.90 ± 3.60        | 70.96±4.17     | 56.33±20.32 | 79.35±5.21  | 82.53±5.54   | 81.50±9.10        | <b>88.52±2.96</b> |
| 13           | 66.08±7.33 | 54.73 ± 1.88 | 67.93 ± 2.28 | 86.36 ± 4.36        | 67.83±5.20     | 67.83±5.20  | 78.49±4.74  | 80.09±4.50   | 78.77±4.33        | <b>84.46±2.80</b> |
| 14           | 86.10±5.35 | 76.66 ± 2.60 | 78.39 ± 1.78 | 95.91 ± 2.47        | 85.54±3.17     | 85.43±8.74  | 94.55±2.26  | 95.28±3.66   | 92.92±5.11        | <b>97.54±0.91</b> |
| 15           | 94.50±1.95 | 93.32 ± 1.00 | 98.58 ± 0.83 | 99.35 ± 0.47        | 94.02±2.90     | 97.13±2.83  | 98.26±1.01  | 98.94±1.06   | 98.89±0.66        | <b>99.26±0.76</b> |
| 16           | 91.36±3.75 | 89.41 ± 1.53 | 90.74 ± 2.92 | 94.29 ± 2.02        | 89.36±3.51     | 91.33±3.20  | 94.82±1.69  | 94.05±4.07   | 97.26±1.37        | <b>98.31±1.48</b> |
| 17           | 91.41±3.83 | 89.81 ± 2.16 | 95.75 ± 1.78 | 95.14 ± 2.61        | 89.21±2.37     | 92.08±3.87  | 96.65±1.96  | 97.54±1.77   | 95.30±5.49        | <b>97.66±1.49</b> |
| 18           | 91.13±2.78 | 90.33 ± 2.19 | 94.03 ± 0.81 | 97.23 ± 1.67        | 90.66±3.82     | 64.21±25.62 | 95.68±2.52  | 97.23 ± 1.67 | 90.66±3.82        | <b>98.80±0.90</b> |
| 19           | 89.90±2.60 | 85.11 ± 1.81 | 81.46 ± 1.40 | 86.18 ± 2.97        | 90.03±1.94     | 67.38±24.16 | 94.89±0.94  | 94.36±1.48   | 94.21±2.12        | <b>95.80±0.74</b> |
| 20           | 92.40±2.76 | 96.38 ± 0.78 | 91.49 ± 2.31 | <b>99.56 ± 0.16</b> | 94.45±2.00     | 75.31±22.05 | 95.69±2.37  | 96.27±2.90   | 97.42±1.69        | 98.90±0.52        |
| 21           | 83.04±9.61 | 87.71 ± 1.64 | 95.23 ± 2.22 | <b>99.69 ± 0.20</b> | 87.86±5.67     | 55.38±36.65 | 93.64±12.16 | 92.61±15.09  | 89.48±11.80       | 99.01±0.66        |
| 22           | 90.53±4.40 | 87.92 ± 1.82 | 94.77 ± 3.57 | <b>99.13 ± 0.66</b> | 91.54±3.50     | 88.84±9.76  | 95.71±2.00  | 97.01±2.54   | 97.14±1.80        | 97.95±1.12        |
| OA (%)       | 85.16±1.18 | 81.26±0.70   | 84.12±1.13   | 89.10±2.83          | 86.36±1.05     | 77.17±9.41  | 91.65±0.85  | 91.19±1.52   | 91.79±2.32        | <b>94.59±0.50</b> |
| $\kappa$ (%) | 81.67±1.39 | 76.98±0.81   | 80.33±1.31   | 86.55±3.32          | 83.03±1.24     | 72.46±10.69 | 89.55±1.05  | 89.04±1.82   | 89.76±2.72        | <b>93.20±0.62</b> |
| AA (%)       | 84.02±1.09 | 80.11±0.72   | 83.81±0.68   | 91.65±1.31          | 84.01±0.83     | 75.03±9.24  | 91.29±0.90  | 91.93±1.60   | 91.28±1.23        | <b>95.01±0.21</b> |

分类性能做出了贡献。具体而言，KSFTB模块通过网络能够自适应地从空间和光谱角度选择并融合适当的感受野特征，提高了不同尺度物体的识别准确率。而TSFTB模块则通过在最相关的token上计算自注意力空间光谱上下文来增强分类性能。这个过程减少了冗余和无关信息的干扰，从而提高了有用特征的集成和利用。当同时应用KSFTB和TSFTB模块时，分类性能得到进一步提升，OA达到96.59%。总之，消融实验的结果表明，KSFTB和TSFTB模块能够同时提高网络在所有四个高光谱数据集上的分类性能，而两个模块的组合展现出协同效应，实现了最佳OA，并证实了这些模块在增强网络分类性能方面的有效性。

(2) 时空频机制的性能贡献。我们进一步探索了KSFTB 中时空频选择机制对分类性能的影响。如表2所示，当单独使用空间或光谱选择分支时，四个数据集的分类性能均下降。这突出了在HSIs中准确识别地面物体时，同时考虑空间和光谱信息的重要性。不仅要自适应地优化空间感受野，还要在适当的尺度上整合光谱特征，因为所需的空间-光谱上下文信息在不同地物覆盖类型中差异很大。例如，在帕维亚大学数据集中，OA分别达到96.09%和96.01%

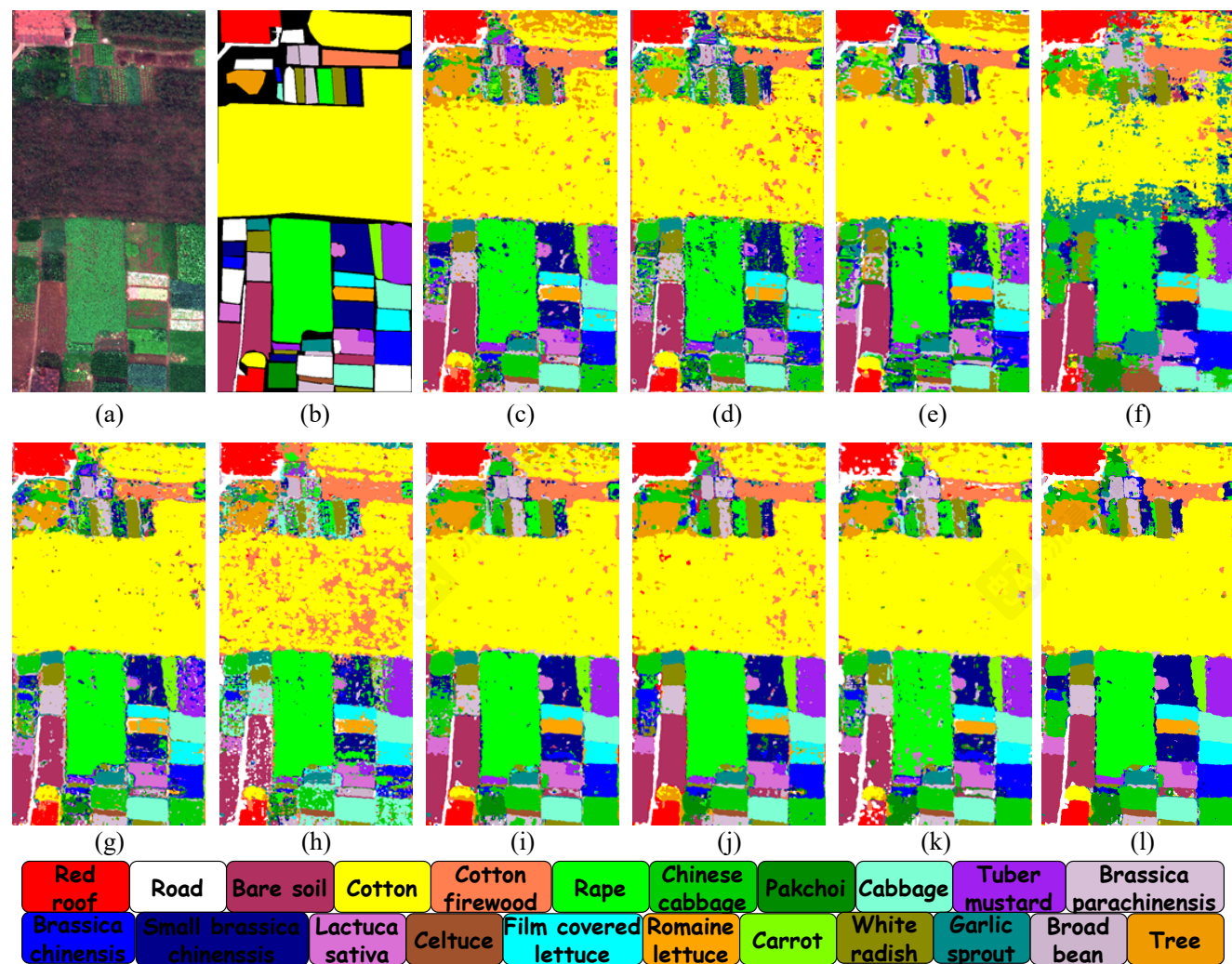
当单独使用空间或光谱分支时。相比之下，当同时应用空间和光谱感受野融合机制时，OA达到96.59%。总之，这项消融实验表明，当同时使用空间和光谱多尺度特征时，可以获得最佳分类性能，从而验证了所提出的KSFTB的有效性。

#### 4.5. 性能比较与分析

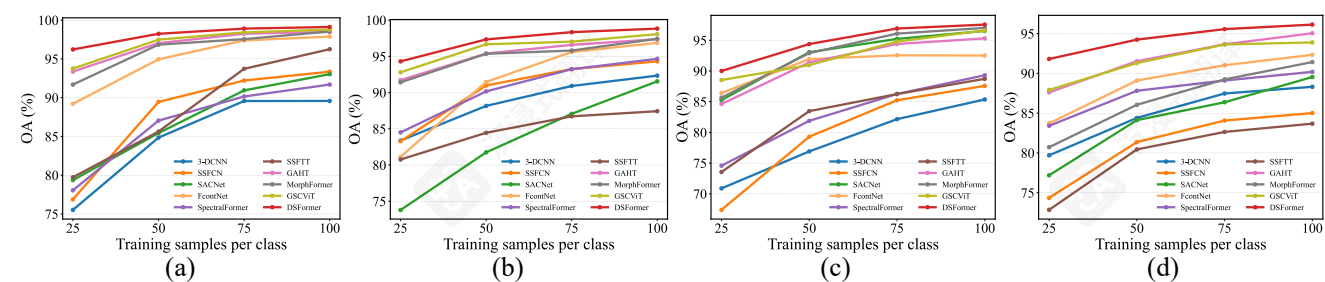
在本小节中，我们将所提出的方法与其他经典和最先进的方法进行比较。以下提供了实验中包含的每种方法的简要介绍。

- 1) 3D-CNN (Ben Hamida等人, 2018): 使用3D-CNN进行空间谱分类。我们遵循原始文章的实现。
- 2) SSFCN (徐等人, 2020): 使用端到端全卷积网络进行空间谱分类。该模型使用Adam优化器进行训练，学习率为  $5e - 4$ ，动量为0.9。
- 3) SACNet (徐等人, 2021): 空间谱分类，一种将自注意力学习与上下文编码相结合的方法。网络使用Adam优化器进行训练，学习率为  $5e - 4$ ，权重衰减为  $5e - 5$ 。
- 4) FcontNet (王等人, 2022a): 基于非局部上下文注意力在空间、波段和尺度上的多尺度分类。基础学习率为0.01，使用多项式调度策略进行调整。模型使用SGD进行优化，动量为0.9，权重衰减为  $1e - 4$ 。





**Figure 10:** Visualization of the classification maps on the Whu-HongHu dataset. (a) False-color image. (b) Ground truth. (c) 3-DCNN. (d) SSFCN. (e) SACNet. (f) FcontNet. (g) SpectralFormer. (h) SSFTT. (i) GAHT. (j) MorphFormer. (k) GSCViT. (l) DSFormer.



**Figure 11:** Comparison of classification performance with different training sample per class: (a) Pavia University dataset. (b) Houston dataset. (c) Indian Pines dataset. (d) Whu-HongHu dataset.

- 5) SpectralFormer (Hong et al., 2022): A Transformer-based architecture that employs sequential learning, focusing on adjacent bands and patches. The model is trained using the Adam optimizer with a learning rate of  $5e-4$ , which decays by a factor of 0.9 after every tenth of the total epochs.
- 6) SSFTT (Sun et al., 2022): Spatial-spectral classification method integrates a CNN with a single-layer

- transformer and a Gaussian distribution-weighted to-tokenization module. The model is trained using the Adam optimizer with a learning rate of  $1e-3$ .
- 7) GAHT (Mei et al., 2022): Spatial-spectral classification through a hierarchical transformer with grouped pixel embedding. The model is trained using the SGD optimizer with a momentum of 0.9, a weight decay of  $1e-4$ , and a learning rate of  $1e-3$ .

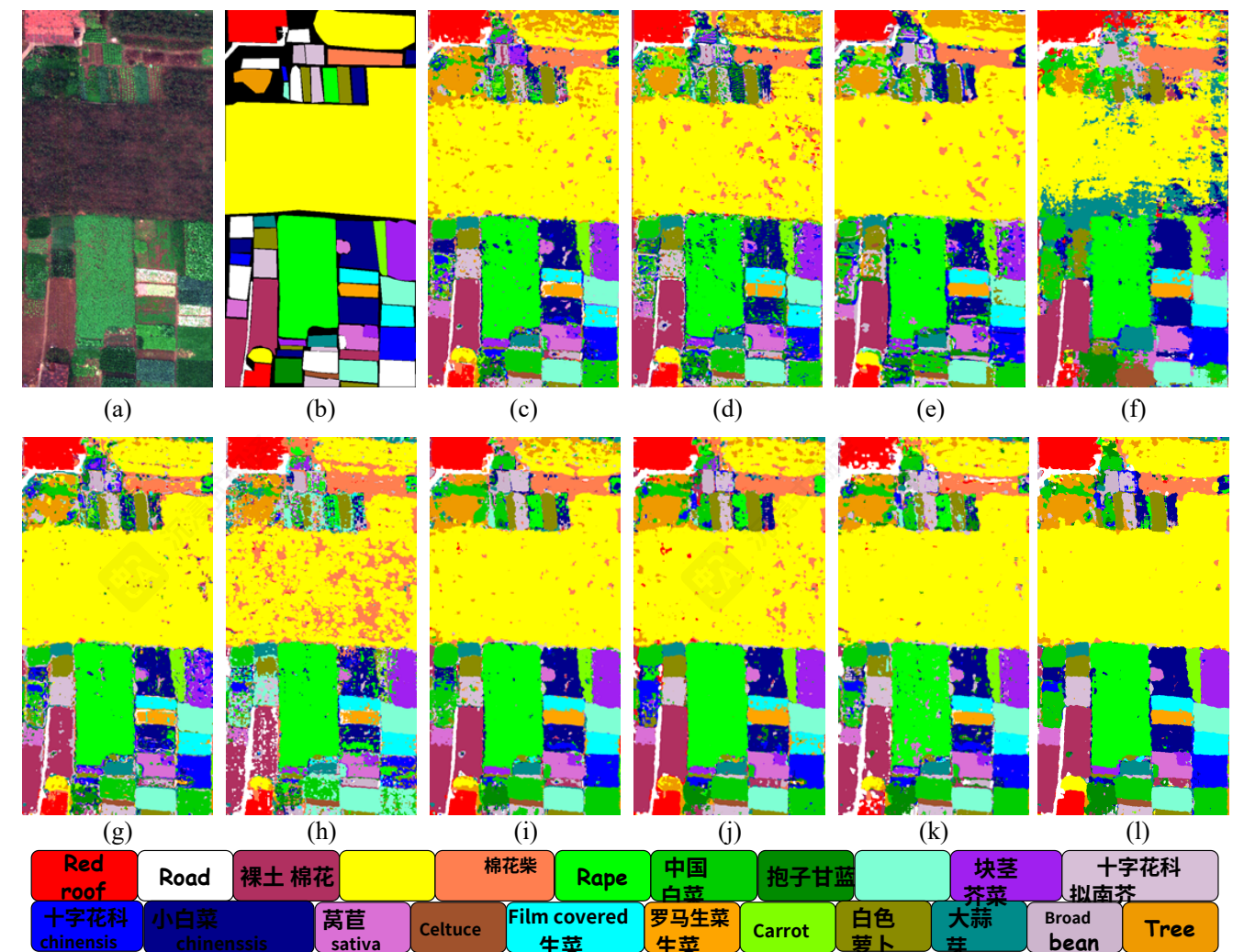


图 10: Whu-HongHu数据集上分类图的可视化。(a) 假彩色图像。(b) 真实情况。(c) 3-DCNN。(d) SSFCN。(e) SACNet。(f) FcontNet。(g) SpectralFormer。(h) SSFTT。(i) GAHT。(j) MorphFormer。(k) GSCViT。(l) DSFormer。

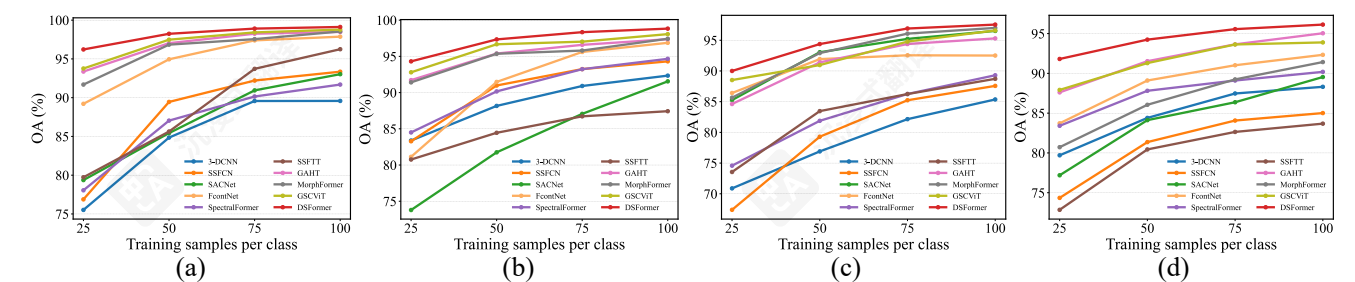


图 11: 不同类别训练样本数量下的分类性能比较: (a) 帕维亚大学数据集。(b) 休斯顿数据集。(c) 印度针叶树数据集。(d) Whu-HongHu数据集。

- 5) SpectralFormer(Hongetal.,2022): 一种基于Transformer的架构, 采用顺序学习, 专注于相邻波段和块。该模型使用Adam优化器进行训练, 学习率为  $5e-4$ , 在总轮数的十分之一后以0.9的因子衰减。
- 6) SSFTT (Sun等人, 2022): 时空频类别方法将卷积神经网络与单层集成

- Transformer和一个高斯分布加权到-tokenization模块。该模型使用Adam优化器以学习率  $1e-3$  进行训练。
- 7) GAHT (梅等人, 2022): 通过具有分组像素嵌入的分层Transformer进行时空频类别。该模型使用SGD优化器进行训练, 动量为0.9, 权重衰减为  $1e-4$ , 学习率为  $1e-3$ 。



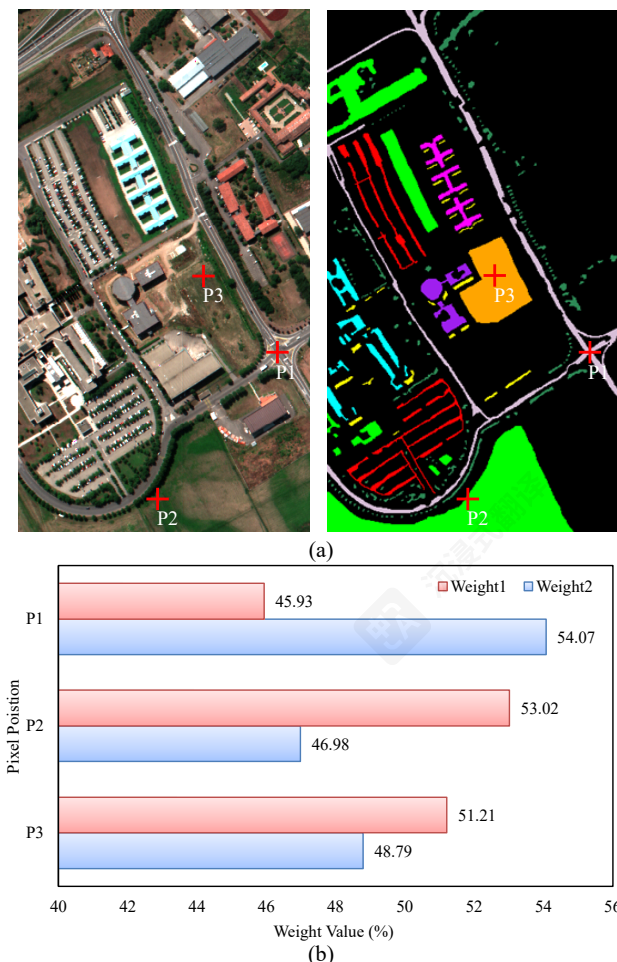
8) MorphFormer (Roy et al., 2023b): A ViT-based spatial-spectral classification method that combines attention mechanisms with morphological operations. The model is trained using the Adam optimizer with a weight decay of  $5e-3$  and a learning rate of  $5e-4$ .

9) GSCViT (Zhao et al., 2024): A lightweight network utilizing groupwise separable convolution and ViT in each feature extraction block. The model is trained with the AdamW optimizer, featuring a learning rate of  $1e-3$  and a weight decay coefficient of 0.05.

**(1) Quantitative Results.** The quantitative results include the accuracy per class, OA,  $\kappa$ , and AA as shown in Table 3 to Table 5, with the best results highlighted in bold. Taking the Pavia University dataset as an example, CNN-based HSI classification methods such as 3D-CNN at the patch level, the classification performance is limited by the size of the input patch, resulting in an OA of only 77.01%. For image-based methods such as SSFCN, although the entire image is used as input, the limited receptive field of the convolutional layer hampers the effective extraction of contextual information, leading to a lower OA of 81.48%. The SACNet method introduces a self-attention learning mechanism and context encoding to establish long-range dependencies, but its performance improvement is restricted by spatial resolution during the self-attention mechanism calculation, yielding an OA of 83.27%. In contrast, FcontNet achieves a relatively higher accuracy with an OA of 90.54% by combining scale, spatial, and channel attention mechanisms for multiscale classification. However, the extensive self-attention computations and multiscale feature design of this method result in high computational resource demands and low operational efficiency.

For transformer-based methods, both SpectralFormer and SSFTT are built on the ViT structure and learn by constructing sequence data from adjacent bands. However, due to redundancy between bands and the lack of spatial context modeling, these methods perform slightly worse than the CNN-based HSI classification methods aforementioned, with an OA of 76.46% and 75.79%, respectively. GAHT builds a hierarchical transformer on the ViT foundation by integrating grouped pixel embeddings, MorphFormer incorporates morphological operations into the ViT attention mechanism, and GSCViT uses groupwise separable convolution with ViT to extract local and global features. All of these methods achieve relatively high accuracy, with OAs of 90.69%, 90.6%, and 93.40%, respectively. However, these approaches overlook the varying contextual ranges required by different types of land cover in the HSI, and the traditional full self-attention mechanism will inevitably includes redundant and irrelevant information during computation. Furthermore, insufficient refinement in capturing the inter-spectral context limits the accuracy of existing transformer-based HSI classification methods.

To address the aforementioned issues, we employ KS-FTB for adaptive receptive field selection and fusion, followed by TSFTB to intelligently choose the most relevant



**Figure 12:** Visualization of the objects with different receptive fields selection weights on the Pavia University dataset. (a) Position of three objects P1: Asphalt, P2: Meadows, and P3: Bare soil. (b) Weights selection values of different receptive fields.

spatial-spectral tokens for self-attention computation. Our method, DSFormer, achieved an optimal OA of 96.59%, surpassing the second-best method by 3.2%. Notably, DSFormer obtained the best results in nearly all per-class accuracy metrics, except for the ninth class, shadows, where it was slightly outperformed by MorphFormer. This may be attributed to the integration of morphological operations within the attention mechanism, which likely provides more clear category priors for shadow recognition. Similarly, our proposed method also demonstrated superior accuracy on other additional datasets, highlighting its advantages and applicability.

**(2) Qualitative Results.** We also present classification maps in Fig. 7-9 to evaluate various methods. Using the Pavia University dataset as a case study, we observed that methods such as 3D-CNN and SSFCN exhibit significant salt-and-pepper noise. In contrast, FcontNet and the latest Transformer-based approaches yield smoother and better results. A particularly noteworthy observation is on the classification of the bricks class, where most existing methods

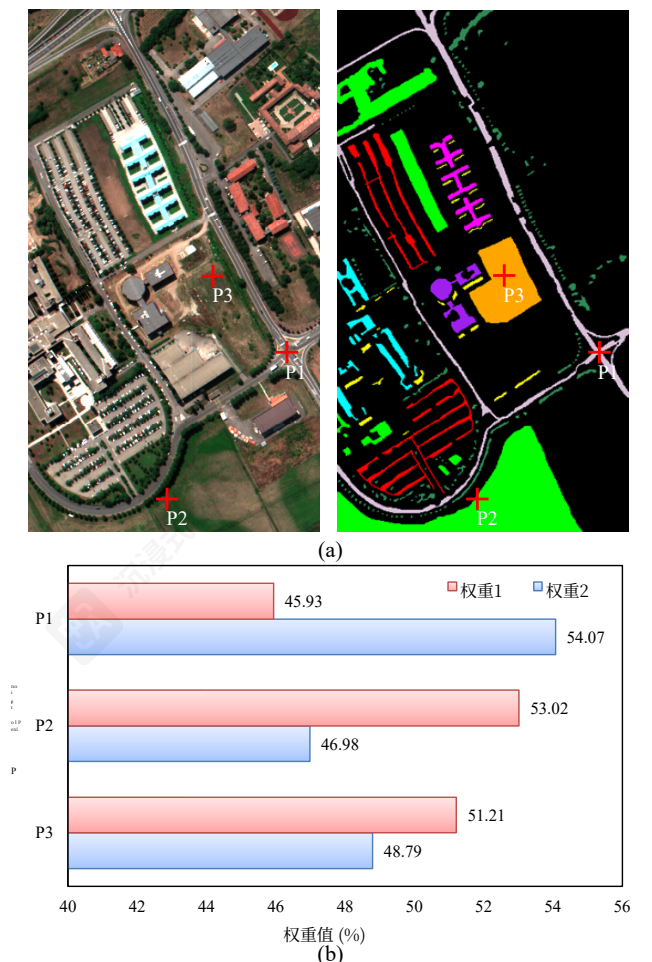
8) MorphFormer (Roy et al., 2023b): 一个基于 ViT 的空间-光谱分类方法，结合了注意力机制与形态学操作。该模型使用 Adam 优化器进行训练，权重衰减为  $5e-3$ ，学习率为  $5e-4$ 。

9) GSCViT (Zhao et al., 2024): 一个利用分组可分离卷积和每个特征提取块中的 ViT 的轻量级网络。该模型使用 AdamW 优化器进行训练，学习率为  $1e-3$ ，权重衰减系数为 0.05。

(1) 定量结果。定量结果包括每类准确率、OA、 $\kappa$  和 AA，如表3至表5所示，最佳结果以粗体突出显示。以帕维亚大学数据集为例，基于CNN的高光谱图像分类方法（如patch级别的3D-CNN）的分类性能受输入 patch 大小的限制，导致OA仅为77.01%。对于基于图像的方法（如SSFCN），尽管整个图像被用作输入，但卷积层的有限感受野阻碍了上下文信息的有效提取，导致OA较低，为81.48%。SACNet方法引入自注意力学习机制和上下文编码以建立长距离依赖，但其性能提升受自注意力机制计算过程中的空间分辨率限制，导致OA为83.27%。相比之下，FcontNet通过结合用于多尺度分类的尺度、空间和通道注意力机制，实现了相对较高的准确率，OA为90.54%。然而，该方法的自注意力计算和多尺度特征设计导致计算资源需求高且运行效率低。

对于基于Transformer的方法，SpectralFormer和SSFTT都基于ViT结构，通过构建相邻波段的数据序列进行学习。然而，由于波段间存在冗余且缺乏空间上下文建模，这些方法的性能略低于上述基于CNN的高光谱图像分类方法，OA值分别为76.46%和75.79%。GAHT在ViT基础上通过集成分组像素嵌入构建了分层Transformer，MorphFormer将形态学操作融入ViT注意力机制，GSCViT则使用与ViT结合的分组可分离卷积来提取局部和全局特征。这些方法均实现了相对较高的准确率，OA值分别为90.69%、90.6%和93.40%。然而，这些方法忽略了高光谱图像中不同地物覆盖类型所需的上下文范围差异，传统的全自注意力机制在计算过程中不可避免地会包含冗余和无关信息。此外，在捕获光谱上下文方面的不足限制了现有基于Transformer的高光谱图像分类方法的准确率。

为解决上述问题，我们采用KS-FTB进行自适应感受野选择与融合，随后使用TSFTB智能选择最相关的



**图12:** 在帕维亚大学数据集上，不同感受野选择权重的物体可视化。(a) 三个物体P1: 沥青、P2: 草地和P3: 裸土的位置。(b) 不同感受野的权重选择值。

时空频标记进行自注意力计算。我们的方法DSFormer达到了96.59%的最优OA，比第二好的方法高3.2%。值得注意的是，DSFormer在几乎所有每类准确率指标中均获得最佳结果，但在第九类“阴影”中稍逊于MorphFormer。这可能是由于注意力机制中集成了形态学操作，这很可能为阴影识别提供了更清晰的类别先验。类似地，我们提出的方法在其他附加数据集上也表现出更高的准确率，突出了其优势和应用性。

(2) 定性结果。我们还展示了图7-9中的分类图来评估各种方法。以帕维亚大学数据集为例，我们观察到3D-CNN和SSFCN等方法表现出明显的椒盐噪声。相比之下，FcontNet和最新的基于Transformer的方法产生了更平滑、更好的结果。一个特别值得注意的观察是砖块类别的分类，其中大多数现有方法



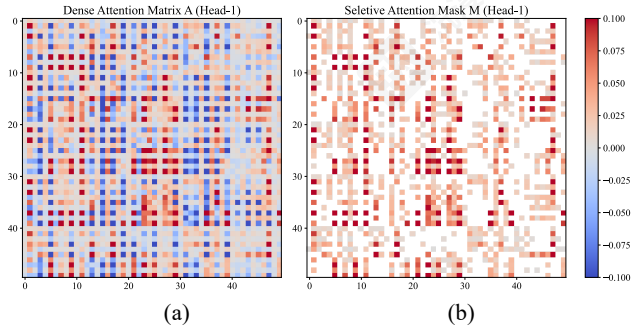
**Table 7**  
Comparison of model parameters (K), inference time (s) and OA (%) across four datasets

| Dataset          | Method             | SpectralFormer | SSFTT   | GAHT    | MorphFormer | GSCViT  | DSFormer |
|------------------|--------------------|----------------|---------|---------|-------------|---------|----------|
| Pavia University | Parameters (K)     | 174.19         | 484.42  | 927.11  | 149.75      | 984.33  | 571.88   |
|                  | Inference time (s) | 44.03          | 8.52    | 15.86   | 26.17       | 14.06   | 39.40    |
|                  | OA (%)             | 76.46          | 75.79   | 90.69   | 90.62       | 93.40   | 96.59    |
| Houston          | Parameters (K)     | 235.95         | 673.74  | 973.58  | 155.19      | 995.22  | 577.90   |
|                  | Inference time (s) | 226.92         | 37.94   | 60.49   | 99.22       | 123.07  | 136.28   |
|                  | OA (%)             | 88.10          | 86.72   | 94.99   | 95.36       | 96.52   | 97.66    |
| Indian Pines     | Parameters (K)     | 352.44         | 931.85  | 830.80  | 199.99      | 1009.62 | 585.20   |
|                  | Inference time (s) | 14.05          | 1.44    | 2.46    | 3.55        | 1.96    | 4.66     |
|                  | OA (%)             | 81.88          | 83.47   | 91.39   | 92.91       | 94.26   | 95.17    |
| Whu-HongHu       | Parameters (K)     | 551.28         | 1254.80 | 1514.97 | 255.99      | 1027.93 | 594.93   |
|                  | Inference time (s) | 274.36         | 45.98   | 63.54   | 83.65       | 49.25   | 108.51   |
|                  | OA (%)             | 86.36          | 77.17   | 91.65   | 91.19       | 91.79   | 94.59    |

exhibit omissions or misclassifications. However, our DSFormer model significantly reduces these errors, producing outputs that are much closer to the ground truth. This advantage has been consistently validated across multiple datasets, highlighting the superior performance of the DSFormer model in handling complex classification tasks.

**(3) Robustness with different training samples.** To assess the stability of the proposed DSFormer method, we conducted extensive experiments on four diverse datasets, each with varying numbers of training samples. Specifically, we randomly selected 25, 50, 75, and 100 samples per class to construct the training sets. Fig. 11 compares the classification performance in these different sample sizes. The results indicate that the classification accuracy of DSFormer is steadily improving as the number of training samples increases, demonstrating its robustness. Furthermore, DSFormer consistently outperforms existing state-of-the-art methods, achieving the highest accuracy across all training sample configurations.

**(4) Computation Cost Analysis.** Given that previous comparative experiments indicate that transformer-based models typically achieve higher accuracy, we performed a comprehensive comparison of these methods regarding parameters, inference time, and OA results, as presented in Table 7. In our performance evaluation, the proposed DSFormer exhibited strong competitiveness and achieved a good balance between precision and efficiency in four benchmark datasets. Regarding model parameters, our DSFormer has fewer parameters than the state-of-the-art methods GAHT and GSCViT. For inference time, DSFormer is faster than SpectralFormer while achieving higher OAs. We believe the efficiency of DSFormer is primarily influenced by the grouping strategy in TSFTB, which generates a large number of tokens for subsequent self-attention calculations. In summary, these results suggest that DSFormer enables a good trade-off between computational resource usage and complex scene perception.



**Figure 13:** Visualization of the alterations in the attention matrix before and after the token selective operation using the Pavia University dataset. (a) The dense attention matrix. (b) The selective attention mask. The attention map is obtained from the first head of MHSA with a group number of 2 and a selection rate of 40%.

#### 4.6. Visualization

**(1) Different Objects of Selection Weights.** To validate the differences in the receptive field requirements for various types of land cover in HSIs, we present visualizations of the selection weights for different objects during receptive field fusion. Fig. 12 (a) presents the locations of three land covers: Asphalt, Meadows and Bare Soil within the Pavia University dataset and their corresponding positions. And Fig. 12 (b) shows the weight values  $W_1$  and  $W_2$  in equation 7 (normalized for visualization) for different receptive fields used in the final fusion within the KSFTB. The weight results show that the weight corresponding to the larger receptive field in P1 is 54.07%, which confirms our previous analysis (see Fig. 1) that asphalt requires more contextual information to perform an accurate classification. In contrast, at locations P2 and P3, both of which exhibit spatial continuity, smaller receptive field characteristics are sufficient for classification.

**(2) Attention Matrix Change of Token Selection.** To further understand the token selection process in TSFTB, we present the changes in the attention matrix before and after the operation (corresponding to equation 13). As shown in Fig. 13, we used the Pavia University dataset with a group number of 2 and a selection rate of 40%. Specifically,

**表7**  
模型参数（K）、推理时间（s）和OA（%）在四个数据集上的比较

| 数据集        | 方法       | SpectralFormer | SSFTT   | GAHT    | MorphFormer | GSCViT  | DSFormer |
|------------|----------|----------------|---------|---------|-------------|---------|----------|
| 帕维亚大学      | 参数 (K)   | 174.19         | 484.42  | 927.11  | 149.75      | 984.33  | 571.88   |
|            | 推理时间 (s) | 44.03          | 8.52    | 15.86   | 26.17       | 14.06   | 39.40    |
|            | OA (%)   | 76.46          | 75.79   | 90.69   | 90.62       | 93.40   | 96.59    |
| 休斯顿        | 参数 (K)   | 235.95         | 673.74  | 973.58  | 155.19      | 995.22  | 577.90   |
|            | 推理时间 (s) | 226.92         | 37.94   | 60.49   | 99.22       | 123.07  | 136.28   |
|            | OA (%)   | 88.10          | 86.72   | 94.99   | 95.36       | 96.52   | 97.66    |
| 印第安松林      | 参数 (K)   | 352.44         | 931.85  | 830.80  | 199.99      | 1009.62 | 585.20   |
|            | 推理时间 (s) | 14.05          | 1.44    | 2.46    | 3.55        | 1.96    | 4.66     |
|            | OA (%)   | 81.88          | 83.47   | 91.39   | 92.91       | 94.26   | 95.17    |
| Whu-HongHu | 参数 (K)   | 551.28         | 1254.80 | 1514.97 | 255.99      | 1027.93 | 594.93   |
|            | 推理时间 (s) | 274.36         | 45.98   | 63.54   | 83.65       | 49.25   | 108.51   |
|            | OA (%)   | 86.36          | 77.17   | 91.65   | 91.19       | 91.79   | 94.59    |

存在遗漏或错误分类。然而，我们的DSFormer模型显著减少了这些错误，产生的输出与真实情况非常接近。这一优势已在多个数据集上得到持续验证，突出了DSFormer模型在处理复杂分类任务方面的优越性能。

(3) 鲁棒性（不同训练样本）。为了评估所提出的DSFormer方法的稳定性，我们在四个不同的数据集上进行了广泛的实验，每个数据集的训练样本数量不同。具体来说，我们为每个类别随机选择了25、50、75和100个样本来构建训练集。图11比较了这些不同样本数量下的分类性能。结果表明，随着训练样本数量的增加，DSFormer的分类准确率稳步提高，展示了其鲁棒性。此外，DSFormer始终优于现有的最先进方法，在所有训练样本配置中均实现了最高准确率。

(4) 计算成本分析。鉴于之前的对比实验表明基于Transformer的模型通常能实现更高的准确率，我们对这些方法在参数、推理时间和OA结果方面进行了全面对比，如表7所示。在我们的性能评估中，所提出的DSFormer表现出强大的竞争力，并在四个基准数据集上实现了精度和效率的良好平衡。在模型参数方面，我们的DSFormer比最先进的方法GAHT和GSCViT的参数更少。在推理时间方面，DSFormer比SpectralFormer更快，同时实现了更高的OA。我们认为DSFormer的效率主要受TSFTB中分组策略的影响，该策略为后续的自注意力计算生成了大量标记。

总之，这些结果表明DSFormer在计算资源使用和复杂场景感知之间实现了良好的权衡。

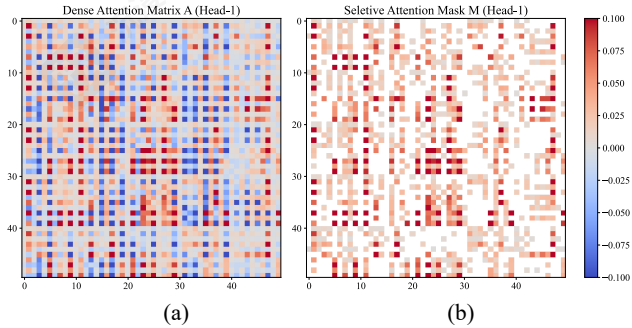


图13：使用帕维亚大学数据集，在token选择性操作前后对注意力矩阵变化的可视化。(a)密集注意力矩阵。(b)选择性注意力掩码。注意力图是从MHSA的第一个头中获得的，组数为2，选择率为40%。

#### 4.6. 可视化

(1) 不同对象的选择权重。为了验证HSI中不同类型地物覆盖在感受野需求上的差异，我们展示了在感受野融合过程中不同对象的选择权重的可视化。图12 (a) 展示了帕维亚大学数据集中三个地物的位置：沥青、草地和裸土及其对应位置。图12 (b) 显示了方程7中  $W_1$  和  $W_2$  的权重值（为可视化而归一化）用于KSFTB最终融合的不同感受野。权重结果表明，P1中对应更大感受野的权重为54.07%，这证实了我们之前的分析（见图1），即沥青需要更多上下文信息来执行准确的分类。相比之下，在位置P2和P3处，两者都表现出空间连续性，较小的感受野特征足以进行分类。

(2) 注意力矩阵变化中的词选择。为了进一步理解TSFTB中的词选择过程，我们展示了操作前后注意力矩阵的变化（对应于公式13）。如图13所示，我们使用了帕维亚大学数据集，分组数量为2，选择率为40%。具体来说，



we selected the dense attention matrix from the first head of the MHSA mechanism. From the figure, it is evident that after the token selection operation, only 40% of the elements in the original dense attention matrix retain their values, while the remaining 60% are set to 0, resulting in a sparse selective attention mask for subsequent effective fusion. Through this process, the relevant information for classification is retained, whereas irrelevant and redundant information is discarded.

## 5. Conclusion

In this article, we introduce a novel network called DSFormer for HSI classification. DSFormer flexibly selects and fuses features across varying receptive fields, achieving contextual modeling of crucial spatial-spectral features through adaptive attention selection and fusion. Recognizing that objects in HSI of varying scales require distinct spatial-spectral contextual ranges, DSFormer incorporates a KS-FTB, which enables the learning of optimal receptive fields by dynamically selecting and integrating multiscale features extracted through multiple dilated convolutions. Furthermore, since the original MHSA may introduce redundant and irrelevant regions and targets that negatively affect HSI classification performance. We develop a TSFTB that realize the adaptive selection of the most valuable spatial-spectral tokens during the context fusion process under a spectral grouping strategy. These selection and fusion mechanisms collectively enable DSFormer to achieve effective visual perception of spatial-spectral contexts within an optimal receptive field. Extensive experiments conducted on four benchmark HSI datasets demonstrate that the proposed DSFormer achieves competitive comprehension performance on accuracy, robustness and inference efficiency compared to existing advanced methods.

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我们从MHSA机制的第一个头中选择了密集注意力矩阵。从图中可以看出，在词选择操作后，原始密集注意力矩阵中只有40%的元素保留了其值，而其余60%被设置为0，从而生成了后续有效融合的稀疏选择性注意力掩码。通过这个过程，分类的相关信息被保留，而不相关和冗余的信息被丢弃。

## 5. 结论

在本文中，我们介绍了一种名为 DSFormer 的新型网络，用于高光谱图像分类。DSFormer 能够灵活地选择和融合不同感受野下的特征，通过自适应注意力选择和融合实现关键时空谱特征的上下文建模。认识到高光谱图像中不同尺度的物体需要不同的时空谱上下文范围，DSFormer 结合了 KS-FTB，该模块通过动态选择和整合通过多次扩张卷积提取的多尺度特征来学习最优感受野。此外，由于原始 MHSA 可能会引入冗余和不相关的区域和目标，从而对高光谱图像分类性能产生负面影响。我们开发了一种 TSFTB，该模块在光谱分组策略下，在上下文融合过程中自适应地选择最有价值的时空谱标记。这些选择和融合机制共同使 DSFormer 能够在最优感受野内有效感知时空谱上下文。在四个高光谱图像基准数据集上进行的广泛实验表明，与现有先进方法相比，所提出的 DSFormer 在准确率、鲁棒性和推理效率方面实现了具有竞争力的理解性能。

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